

MODULE 4: BLOOD AND IMMUNITY MODULE

EXIT EXAM PREPARATION MODULE

Friday, January 29, 2015

SECTION I: ANATOMY, HISTOLOGY AND
EMBRYOLOGY OF LYMPHOID ORGANS

*DESCRIPTIVE CONSIDERATION OF
ANATOMICALLY RELEVANT SITES*

- Broad classification of Immune system: Acquired and Innate.
- T-cell, B-cell immunity and Blood borne defenses of Innate Immunity ultimately result in death of the dangerous pathogens while physical barriers prevents the entry of infection.
- Cells of the immune system are distributed in the blood, lymph, connective and epithelial tissues.
- Lymphoid nodules are small spherical constellations of immune cells found in connective tissues and inside several organs.
- Lymphoid organs include lymph nodes, spleen, the thymus and the bone marrow.
- Mucosa associated lymphoid tissue (MALT) are diffuse lymphoid nodules and isolated cells in various functional systems of the body. These include:
 - The digestive system: Tonsils, Peyer's Patches and Appendix
 - The Respiratory System: Six Nodes
 - The Reproductive System: Three Nodes
 - Urinary System: Two type of nodes and Thoracic Duct (Posterior Abdominal Wall)

LYMPHOID ORGANS

- The lymphoid organs are divided into two: primary and secondary.
- Primary Lymphoid organs are sites of lymphocyte production and maturation.
- Secondary Lymphoid organs are sites of lymphocyte activation.
- Primary Lymphoid organs are thymus and bone marrow.
- Secondary lymphoid organs are the spleen, lymph nodes and MALT.
- Based on encapsulation by the connective tissue, lymphoid organs are classified into three categories:
 - *Completely encapsulation:* Thymus, Spleen and Lymph Nodes
 - *Partial/Incomplete encapsulation:* Tonsils and Adenoids
 - *No encapsulation:* MALT, Appendix and Temporary aggregation of lymphocytes and immune cells

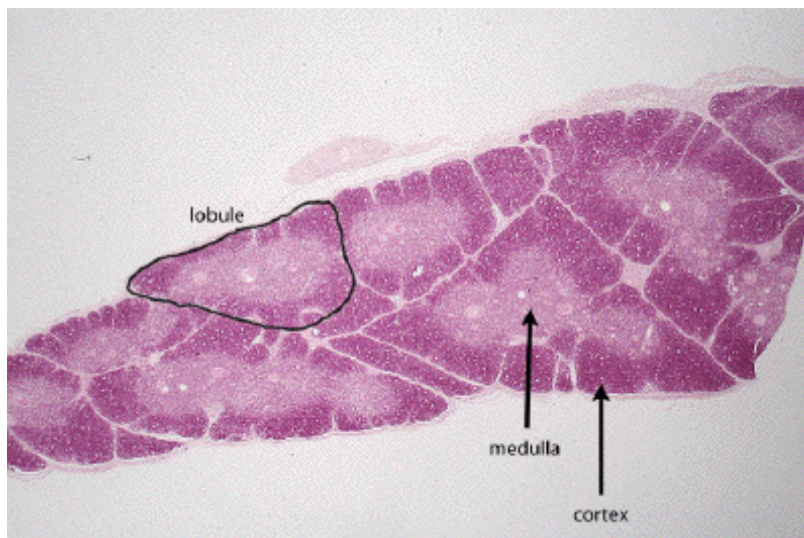
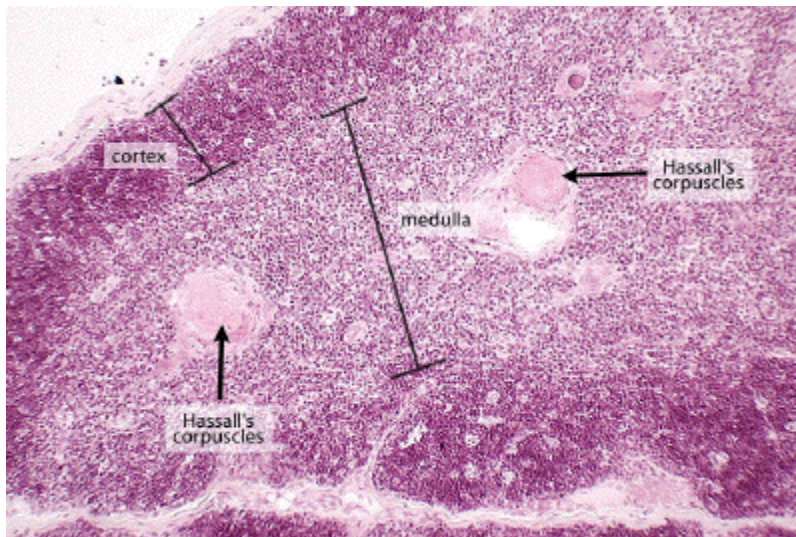
PRIMARY LYMPHOID ORGANS

Thymus

- It is a bi-lobed structure of the anterior part of the superior mediastinum that grows until puberty. From that point onwards, it involutes.
- Arterial supply of the thymus is derived from the inferior thyroid arteries, anterior mediastinal branch of the internal thoracic artery and anterior intercostal arteries.
- Thymic veins drain into left brachiocephalic, internal thoracic and inferior thyroid veins.
- Lymphatic drainage of the thymus primarily involves parasternal, tracheobronchial and brachiocephalic lymph nodes.
- The thymus is divided into **cortex and medulla incompletely** by connective tissue (fibrous) septa.
- The thymus is derived from the **endoderm and mesoderm**.
- The endodermal component is the **epithelial reticular cells** and the mesodermal component are the **lymphocytes and the perivascular loose connective tissue**.
- These partitioning and stratification are blurred or somehow indistinguishable as one ages.
- The thymic **cortex** is the site where **immature thymocytes** are localized.
- In addition to thymocytes, it contains epithelial reticular cells which are characterized as being pale cells with long cytoplasmic processes **encircling thymocytes**. They are large, ovoid with pale-staining nucleus containing nucleolus.
- These cells produce **thymosin, serum thymic factor, and thymopoietin**. These modulate the transformation of immature T-cells into competent T-cells.
- These cells contain **tonofilaments essential for the formation of desmosomal contacts with each other**.
- The **Blood Thymus Barrier** exists in the thymic cortex and its main purpose is to **prevent the exposure of immature T-cells to blood antigens**.
- Constituents of the Barrier include:
 - Thymic **capillaries endothelium and their basal laminae**
 - **Perivascular connective tissue and cells**

➤ Epithelial reticular cells and their basal laminae

- Thymocytes are cells that become naïve T-cells as soon as they complete their thymic modulation.
- Thymocytes are surrounded by the process of the reticular cells.
- As these thymocytes develop and differentiate, they express receptors that help them recognize foreign antigens.
- The developing thymocytes migrate towards the medulla. Most of them die before reaching there.
- Those that die are engulfed by macrophages and this happens most notably in the case of self-antigen recognizing T-cells. These types of T-cells are forced into apoptosis.
- The mature T-cells enter secondary lymphoid organs via the vascular system.



- Thymic medulla is, as demonstrated in the slide above, slightly less cellular and more heterogeneous than the cortex.
- The reason this part of the thymus stains light is because it contains many epithelial reticular cells and naïve T cells.
- Naïve T-cells enter circulation via efferent lymphatic vessels and venules.
- The thymic medulla contains the unique marker of identification that we know as the Hassall Corpuscle.
- The Hassall Corpuscle is a whorl-like accretion of the reticular cells that often displays concentric keratinization.
- For some unknown reason, these corpuscles increase with age.
- In the medulla, various hormones are introduced and to some extent elaborated. Some of them are listed below:
 - Thymic Stromal Lymphopoietin: produced by reticular cells and facilitate dendritic maturation.
 - Thymosin α 1: increases the interferon, the lymphokine, and the MIF (migration inhibitory factor) production. Also increases immunity against viruses, fungi and tumors.
 - Thymosin β 4: stimulates the release of LH-RH and induces the expression of receptors in the bone marrow of healthy mice and in mice without a thymus, as well as in vivo in the thymocytes of mice with a depressed immune system.
 - Thymopoietin helps the intrathymic lymphocytes to differentiate amongst the bone marrow cells.
- DiGeorge Syndrome is a condition that is characterized by congenital thymic aplasia. There is a congenital absence of thymus and parathyroid glands. There is an abnormal development of the third and fourth pharyngeal pouches. It produces abnormal cell-mediated immunity with intact humoral immunity. Tetany (carpal pedal) due to lowered calcium levels or death secondary to severe infections is not a rarity.

Bone Marrow

- The marrow is located within the medullary cavity between the trabeculae of the spongy bone.
- The bone marrow is the site of blood cell production and maturation. It is also the site for the production and maturation of B-cells.
- There are two types of bone-marrow in the body: Red and Yellow bone marrow.
- Red marrow is hemopoietic and located in the epiphyses of long bones and in short bones, flat and irregular bones.
- Active bone marrow in the adults are located in the ribs, vertebrae, pelvis and shoulder girdle.
- Yellow marrow is hemopoietic when the situation warrants it. Otherwise, it is filled with fat and is located in the long bones.
- Yellow marrow can revert to red marrow in either of the two ways:
 - By stem cell invasion of the marrow or
 - By extension of hemopoietic cells into the inactive marrow.
- Bone marrow becomes an organ of hemopoiesis after 20 weeks of gestation and becomes the main hematopoietic organ by the time of birth.
- The red marrow is composed of a stroma, irregular sinusoids and hemopoietic cells.
- The stroma is composed of stromal cells such as macrophages, adventitial reticular cells, fibroblasts and endothelial cells.
- A key component of the stroma are the adventitial reticular cells.
 - These cells extend processes into the substance of the hemopoietic cords and aid in the development of precursor blood cells.
 - They secrete various cytokines, such as CSFs, IL-5, IL-7, that aid in the development of precursor cells into mature blood cells.
- The sinusoids are vessels lined with attenuated endothelia and are associated with reticular fibers on their extravascular surface.
 - The sinusoids of the bone marrow derive from vessels that have just nourished the cortico-medullary region of the bones.
 - Their walls consist of endothelial lining, basal lamina and incomplete covering of adventitial cells.

- Newly formed blood cells migrate transcellularly in to the sinusoidal lumen. This happens by the fusion of the luminal plasma lemma of the endothelium and the ad luminal plasma lemma of the endothelium, literally piercing it.
- The hemopoietic compartment are characterized as being more of “cords” than cell like.
 - There exist cell clusters for erythrocytes, megakaryocytes and granulocytes.
 - Clusters for erythrocytes and megakaryocytes are located adjacent to the sinusoidal walls while that of granulocytes are found at quite a distance to it. As a result, granulocytes have to migrate to reach the sinusoidal wall.
 - Each cluster containing erythrocytes contains a macrophage

SECONDARY LYMPHOID ORGANS

Spleen

- The spleen belongs to the reticuloendothelial system.
- Its main function is clearing the blood off worn-out red blood cells, storing erythrocytes in the case of hemorrhage, and site of B and T-cell activation as well as proliferation.
- The spleen is the largest lymphoid organ in the body.
- It is ovoid in shape and purplish pulpy mass.
- The spleen is located between the 9th and 11th rib.
- Its long axis lies roughly parallel to the 10th rib.
- It is located in the Left Upper Quadrant at the hypochondrium.
- Its inferior pole lies in the mid-axillary line.
- The Hilum of the spleen serves as a conduit for vessels entering the spleen.
 - It is capable of marked expansion and contraction.
- Has three borders:
 - Anterior (which is sharp and notched like that of the superior border)
 - Inferior (which is rounded)
 - Superior border
- It is an organ that receives arterial supply from the splenic artery branches (which could be up to 6 branches). Venous drainage is by the corresponding vein.
- The splenic veins join the superior mesenteric veins to form the hepatic portal vein.
- An interesting clinical note is that the spleen could enlarge to a considerable size during portal hypertension and trauma to the anterolateral abdomen.
- Rupture of the spleen is a dangerous condition that can result in hemorrhage and subsequent shock.
- The spleen is suspended securely by two structures:
 - Gastrosplenic omentum: conveys short gastric vessels and left gastroepiploic vessels. It also attaches the spleen to the greater curvature of the stomach.
 - Splenicorenal ligament: conveys the splenic vessels to the hilum of the spleen and contains the tail of the pancreas. This also attaches the spleen to the left kidney.

- Splenic lymphatic vessels finally deliver their contents to the celiac nodes via the pancreaticosplenic nodes.
- The innervation for this lymphoid organ arises from the celiac plexus.
- The spleen is enclosed by a thin fibrous capsule which consists of dense irregular connective tissue.
- This thin fibrous capsule extends trabeculae into the substance of the spleen bringing with trabecular vessels to supply the splenic pulp.
 - The trabeculae of the fibrous capsule contain myofibroblasts which aid in the expulsion of stored RBCs in the spleen.
 - Lymphatic vessels originate near the trabeculae.
 - No lymphatic vessels exist in the pulp.
- The splenic pulp consists of two functional units: Red and White pulp.
- Red pulp consists of cords i.e. cords of Billoth and splenic sinuses.
 - Splenic sinuses are lined by rod-shaped endothelial cells with few contact point between each other.
 - This feature helps in the entrance of RBCs after percolating in the cords of the red pulp.
 - This also helps macrophages and dendritic cells extend their process between the spaces to sense foreign antigens present in the circulation.
 - These vessels are tied perpendicular to their long axis by an incomplete strand of basal lamina.
 - Splenic cords consist of a loose meshwork of reticular fibers and cells containing RBCs, macrophages, dendritic cells, lymphocytes, plasma cells and granulocytes.
 - These cells help in identifying antigens and mounting response towards them.
 - They also acquire iron from worn-out or old RBCs
 - Spleen has two types of circulations:
 - *Open circulation theory*: Once the splenic artery reaches the capsule, it branches to capsular and trabecular components. The trabecular component, as it passes through the white pulp, becomes the central artery. This also supplies marginal sinuses around the white pulp. The penicillar arterioles extending from the central artery continue as arterial capillaries. These might be sheathed by T-lymphocytes centrally and B-cells more

peripherally (aka PALS). When these sheathed capillaries spill into the reticular fibers of the cords, the blood would percolate the space until it joins the sinuses. This is what is called *open circulation*.

- *Closed circulation theory*: This occurs only in some organisms such as the rat and pigs. It occurs when the blood from the capillaries goes directly to the **splenic sinusoids/sinuses**.
- White pulp is described as lymphoid aggregation around a central artery.
 - It has a central **PALS**, a peripheral white pulp and germinal center.
 - The PALS consists of T-cells that form a cylindrical enclosure around the central artery throughout its whole course.
 - The peripheral white pulp consists chiefly of B-cells.
 - **Germinal center are sites within secondary lymphoid organs where mature B lymphocytes proliferate, differentiate, and mutate their antibody genes (through somatic hypermutation), and switch the class of their antibodies (for example from IgM to IgG) during a normal immune response to an infection.**
 - In cross-sections of the PALS, it resembles lymphatic nodules.
 - In humans, germinal centers develop within 24 hours after antigen exposure.
 - These become what are splenic nodules (**Malpighian Corpuscles**).
- The spleen has two broad functions:
 - Hemopoietic
 - Erythropoiesis in fetal life
 - Destruction of worn-out RBCs
 - Iron reclamation and hemoglobin break-down
 - Storage of erythrocytes
 - Storage of platelets
 - Immunological functions
 - Interface between APCs and effector response from lymphocytes
 - Site of B and T-cell activation

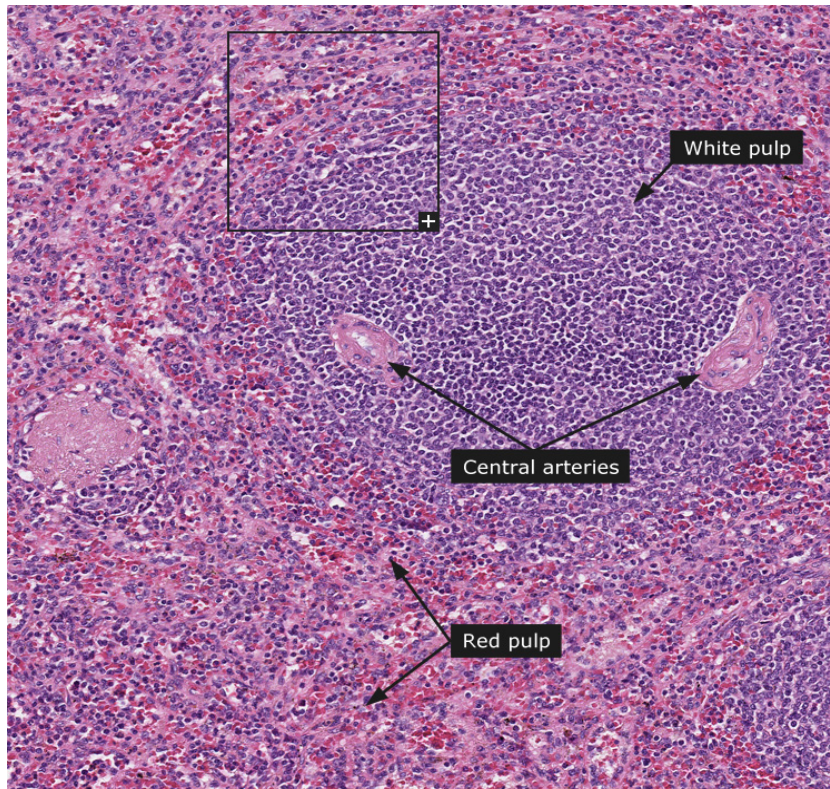


Figure 3: Diagrammatic Representation of the General Structure of the Spleen.

- Appreciate the appearance of the red pulp, so named due to its reddish appearance and pulp-like consistency in fresh histological preparation.
- Appreciate the white pulp, so named to its whitish appearance in fresh histological preparations.
- The white pulp possess a central artery named due to the vessels encapsulation by the lymphatic tissue.
- Due to intermittent contraction and relaxation of the central artery, it is rather eccentrically placed as would fit a cuff around the wrist.

Lymph Nodes

- Lymph nodes are the **most diffuse and largest total of the lymphoid organs** in the human body.
- It is the site where **lymph is filtered and cleared**.
- Lymph nodes are encapsulated and receive lymph brought to them by lymphatic vessels.
- In the end, lymph nodes transfer their contents into the venous blood.
- One can broadly classify the various lymph nodes in the body into two groups
 - Superficial Lymph Nodes:
 - Cervical,
 - Axillary and
 - Inguinal Lymph Nodes
 - Deep Lymph Nodes:
 - Tracheobronchial,
 - Aortic and
 - Iliac Lymph Nodes
- All lymph nodes in the right superior one-fourth of the body drain into the right lymphatic duct.
- All the rest drain into the thoracic duct.
- Both, the right lymphatic duct and thoracic duct, enter the venous circulation in sites called the venous angles.
- Venous angles are formed by the union of the two with their respective Internal Jugular vein.
- There are two types of lymphatic vessels serve the lymph nodes
 - Afferent lymphatic vessels: Less than 10% of lymph enters the lymph nodes through this way. Enters at the convex region of the node.
 - Efferent lymphatic vessels: they take lymph away from the nodes unlike the afferents. These vessels leave through hilum together with arteries and veins.
- Blood vessels convey antigen-primed lymphocytes to the nodes. Here they undergo proliferation and differentiation.
- Supporting elements of the lymph nodes are the following:
 - **Capsule:** Consists of dense connective tissue and covers the lymph nodes
 - **Trabeculae:** it is an invagination of the capsule, consisting of dense connective tissue, into the substance of the node.

- **Reticular tissue**: consists of reticular cells and reticular fibers. It forms a meshwork of supporting structure of the node.
 - Cells of the reticular network are composed of three cells
 - Reticular cells: synthesize and secrete collagen type III and associated ground substance that forms the stroma.
 - Dendritic cells (DC): bone marrow derived APCs that present various protein antigens and also MHC I as well as MHC II
 - Macrophages: both phagocytic and APCs that express MHC I, MHC II and costimulatory molecules.
 - Follicular Dendritic Cells (FDCs): dendritic cells that interdigitate between B-lymphocytes of germinal centers and lack MHC II.
- General architecture of the node consist of cortex, paracortex and medulla.
 - Cortex: Consists of the superficial and deep cortex
 - Superficial cortex: contains nodules and if it is stimulated contains germinal centers.
 - Deep cortex (para-cortex): a portion between the superficial cortex and medulla. It is devoid of any lymphatic nodules.
 - It is also one of the thymus-dependent zone.
 - Contains vessels lined by cuboidal or columnar epithelial cells known as High Endothelial Venules (HEV)
 - Medulla: consists of medullary cords and sinuses
 - Medullary cords: contain mostly B-lymphocytes.
 - Medullary sinuses: drain into efferent lymphatic vessels.

- Form a ring of lymphoid tissue at the entrance of the oropharynx.
- There are three of them at the entrance:
 - Pharyngeal tonsils: also called adenoids. It is located on the roof of the pharynx.
 - Palatine tonsils: located on either side of the pharynx between the palatoglossal and palatopharyngeal arches.
 - Lingual tonsils: found at the base of the tongue
- Tonsils consist of squamous epithelium on the surface that dips into the underlying connective tissue in several places forming tonsillar crypts.
- They don't possess afferent lymphatic vessels.

Peyer's Patch and Gut-associated Lymphatic Tissue (GALT)

- Found in the ileum.
- There also exist numerous solitary nodules along the way between the small and large intestine.

Embryogenesis of Lymphoid Organs: Brief Review

Primary Lymphoid Organs

Thymus

- The thymus is derived from the endoderm of third pharyngeal arch.
- The third pharyngeal pouch expands and develops a solid, dorsal bulbar part and a hollow, elongate ventral part.
- The epithelium of the elongate ventral parts of the pouch proliferates, obliterating their cavities.
- These come together in the median plane to form the thymus.
- Up to the sixth week, the thymus is purely epithelial.

Bone marrow

- The production of blood cells in bone marrow begins roughly 4-5 months after conception.
- Stem cells immigrate from the liver into the bone marrow, where the "microenvironment" is decisive for the development of stem cells.
- This stroma consists of endothelial cells, fat cells, osteoblasts and fibrocytes.
- Macrophages also colonize the stroma, but they stem from hematopoietic stem cells.

Secondary Lymphoid Organs

Spleen

- This organ is derived from a mass of mesenchymal cells located between the layers of the dorsal mesogastrium.
- The spleen, a vascular lymphatic organ, begins to develop during the fifth week but does not acquire its characteristic shape until early in the fetal period.
- The spleen is lobulated in the fetus, but the lobules normally disappear before birth.
- The notches in the superior border of the adult spleen are remnants of the grooves that separated the fetal lobules.
- As the stomach rotates, the left surface of the mesogastrium fuses with the peritoneum over the left kidney.

- This fusion explains the dorsal attachment of the splenorenal ligament and why the adult splenic artery, the largest branch of the celiac trunk, follows a tortuous course posterior to the omental bursa and anterior to the left kidney.

Tonsils

- Although the second pharyngeal pouch is largely obliterated as the palatine tonsil develops, part of the cavity of this pouch remains as the tonsillar sinus or fossa.
- The endoderm of the second pouch proliferates and grows into the underlying mesenchyme.
- The central parts of these buds break down, forming tonsillar crypts (pit-like depressions).
- The pouch endoderm forms the surface epithelium and the lining of the tonsillar crypts.
- At approximately 20 weeks, the mesenchyme around the crypts differentiates into lymphoid tissue, which soon organizes into the lymphatic nodules of the palatine tonsil.

*BONE MARROW, THYMUS, SPLEEN
AND MALT*

- **Bone Marrow:** Bone marrow and adipocytes are found in the medullary canals of long bones and in the cavities of cancellous bones. Based on their appearance at gross examination can be divided to red bone marrow (blood forming component, and site where old RBC's are phagocytosed and Hg is stored) and yellow bone marrow (filled with adipocytes). {In the new born essentially all the marrow is red; when required the yellow marrow in adults can revert to red marrow}.
- **Red bone marrow:** composed of a *stroma*, *hemopoietic cords (islands of cells)*, and *sinusoidal capillaries*. Stroma composed of reticular cells (specialized fibroblasts), and reticular fibers supporting blood forming cells and macrophages. Sinusoids are formed by a thin layer of endothelial cells, through gaps in these endothelial cells mature blood cells exit the marrow (RBC's-through pressure gradient, WBC's-through diapedesis, Platelet-through thin processes)
- **Spleen:** peritoneum covers the capsule (dense irregular CT) of the spleen, which sends trabeculae into the substance of the spleen to form a supportive framework. Divided into 1) *red pulp* and 2) *white pulp* (has lymphoid elements) separated by the 3) *marginal zone*.
 - **Red pulp of the spleen** is composed of an interconnected network of *sinusoids* (are lined by long fusiform endothelial cells separated by relatively large blood containing intercellular spaces) supported by a loose type of reticular tissue (splenic cords of Billroth). Splenic cords contain plasma cells, stellate reticular cells, blood cells, and macrophages enmeshed within the spaces of the reticular fiber network. Processes of the macrophages enter the lumina of the sinusoids through the spaces between the endothelial cells. {Responsible for phagocytic functions associated with spleen}
 - **White pulp of the spleen** includes **all** of the organ's lymphoid tissue (diffuse and nodular), such as **lymphoid nodules** (mostly B cells) and **PALS** (mostly T cells and found around the central arteries). It also contains macrophages and other **APC's** (antigen presenting cells).
 - **The marginal zone of the spleen** (a sinusoidal region between the red and white pulps at the periphery of the PALS) receives blood from capillary loops derived from the central artery and thus is *the first site where blood contacts the splenic parenchyma* is richly supplied by

avidly phagocytic macrophages and other APCs, is the region where circulating T and B lymphocytes enter the spleen before becoming segregated to their specific locations within the organ and **where interdigitating dendritic cells are able to display their MHC-epitope complex for recognition by T cells**

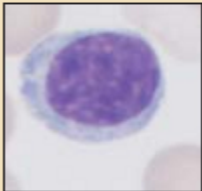
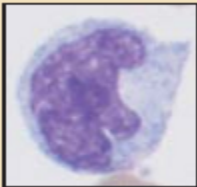
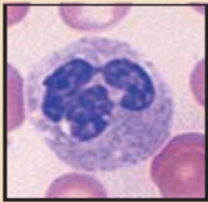
- **Vascularization of the spleen** is derived from the **splenic artery**, which enters the hilum and gives rise to *trabecular arteries*-which leaves the trabeculae become invested by a periarterial lymphatic sheath (PALS) and are known as *central arteries*- these arteries branch but maintain their lymphatic sheath until they leave the white pulp to become many straight *penicillar arteries*- *these enter the red pulp and have 3 regions: pulp arterioles, macrophage-sheathed arterioles, and terminal arterial capillaries*. These last named vessels either drain directly into the splenic sinusoids (closed circulation) or terminate as open-ended vessels within the splenic cords of the red pulp (open circulation).

{The spleen has no **afferent lymphatic vessels; hence has no function in filtration**}

- **MALT** (70 % of body's immune cells) is diffuse lymphoid tissue found in the mucosa of the G.I and Resp. Tract and consists of two major types, bronchus-associated lymphoid tissue (BALT) and gut associated lymphoid tissue (GALT). Both types possess lymphoid nodules that are isolated from one another (except in the case of Peyer patches.) GALT includes tonsils, Peyer's patches, and appendix.
- **Peyer patches** are aggregates of lymphoid nodules found in the ileum. Their surface is covered by **M-cells**: cells with microfolds over their surface that sample Ag's in the intestinal lumen, and have porous basement membranes. Antigenic material bound to the apical surface of M cells is rapidly translocated via transcytosis from the intestinal lumen to APCs and lymphocytes located in their Peyer's patches.
- **Tonsils**: are aggregates of lymphoid tissue, which sometimes lack a capsule. All tonsils are in the upper section of the digestive tract, lying beneath but in contact with the epithelium.
 - 1) **Palatine tonsils**: possess crypts, (deep invaginations of the stratified squamous epithelium covering of the tonsils), frequently containing debris. Possess primary and secondary lymphoid nodules, and are separated from subjacent structures by a connective tissue capsule.

- 2) The pharyngeal tonsil is a single tonsil in the posterior wall of the nasopharynx. It is covered by a pseudostratified ciliated columnar epithelium, and has longitudinal infoldings instead of crypts.
- 3) Lingual tonsil. On the dorsum of the posterior third of the tongue and is covered by a stratified squamous nonkeratinized epithelium. Has deep crypts, which frequently contain cellular debris. Ducts of the posterior mucous glands of the tongue often open into the base of these crypts

*CELLS OF THE IMMUNE SYSTEM,
BLOOD BUFFER SYTEM,JAUNDICE AND
PROTEIN BIOMARKERS IN CLINICAL
DIAGNOSIS*

Cell type	Principal function(s)
Lymphocytes: B lymphocytes; T lymphocytes; natural killer cells  <i>Blood lymphocyte</i>	Specific recognition of antigens: B lymphocytes: mediators of humoral immunity T lymphocytes: mediators of cell-mediated immunity Natural killer cells: cells of innate immunity
Antigen-presenting cells: dendritic cells; macrophages; follicular dendritic cells  <i>Dendritic cell</i>  <i>Blood monocyte</i>	Capture of antigens for display to lymphocytes: Dendritic cells: initiation of T cell responses Macrophages: initiation and effector phase of cell-mediated immunity Follicular dendritic cells: display of antigens to B lymphocytes in humoral immune responses
Effector cells: T lymphocytes; macrophages; granulocytes  <i>Neutrophil</i>	Elimination of antigens: T lymphocytes: helper T cells and cytotoxic T lymphocytes Macrophages and monocytes: cells of the mononuclear-phagocyte system Granulocytes: neutrophils, eosinophils

Cells of the immune system

Adaptive immune cells

- Lymphocytes - are the only cells that produce specific receptors for antigens and are thus the key mediators of adaptive immunity
- Lymphocytes: all lymphocyte progenitor cells originate in the bone marrow. Some of these lymphocytes migrate to the thymus, where they acquire the full attributes of T lymphocytes. Other bone marrow lymphocytes differentiate into B lymphocytes in the bone marrow.
- B lymphocyte - main function is generation of, antibody producing terminal cells (plasma cells).
- Helper T cells: they help B lymphocytes to produce antibodies and help phagocytes to destroy ingested microbes.
- Regulatory T lymphocytes. Belong to a special subset that functions to prevent or limit immune responses
- CD8⁺ T lymphocytes: are called cytotoxic, or cytolytic, T lymphocytes (CTLs) because they kill ("lyse") cells harboring intracellular microbes
- Natural killer cells: lacks T and B cell marker. Attacks virus-infected cells and cancer cells without previous stimulation.
- When naive lymphocytes recognize microbial antigens and also receive additional signals induced by microbes, the antigen-specific lymphocytes proliferate and differentiate into effector cells and memory cells
- Effector cells are the differentiated progeny of naive cells that have the ability to produce molecules that function to eliminate antigen E.g. - plasma cells.
- Memory T cells - are progeny of T cells that have proliferated in response to antigen which are long-lived and functionally inactive and circulate for months or years, ready to rapidly response to repeat exposures to the same microbe.

Antigen presenting cells

- Dendritic cells- specialized antigen-presenting cells (APCs) located in the epithelium of (skin, GI tract, and respiratory tract) that capture antigens, transport them to peripheral lymphoid tissues, and display them to lymphocytes. Dendritic cells are capable of ingesting virus infected cells and displaying viral antigens bound to class I MHC molecules to CD8⁺ T lymphocytes (cross-presenting).
- Follicular dendritic cell (FDC) specialized DC's resides in the germinal centers of lymphoid follicles in the peripheral lymphoid organs and displays

antigens that stimulate the differentiation of B cells in the follicles. They do not present antigen.

- Effector cells- The cells that eliminate microbes are called effector cells and consist of lymphocytes and other leukocytes (macrophages and granulocytes)
- Macrophages - Part of the mononuclear phagocytic system. Function in phagocytosis and digestion of protozoa and virus and senescent cells
- Granulocytes (neutrophils) - (also called polymorphonuclear leukocytes [PMNs]) the production of neutrophils is stimulated by cytokines, known as colony-stimulating factors. Neutrophils are the first cell type to respond to most infections, particularly bacterial and fungal infections.

Other cells of innate immune system

- Eosinophils- Defense against parasitic helminthes; modulation of inflammatory processes
- Basophils - function by Release of histamine and other inflammation mediators. Specific granules containing histamine and heparin

The blood buffer system

- The hemoglobin in the cells is an excellent acid-base buffer (as is true of most proteins), so the RBCs are responsible for most of the acid base buffering power of whole blood.
- RBC's contain a large quantity of carbonic anhydrase which catalyze reversible reaction between carbon dioxide (CO₂) and water to form carbonic acid (H₂CO₃), increasing the rate of this reaction 5000 fold.
- This reaction makes it possible for the water of the blood to transport enormous quantities of CO₂ in the form of bicarbonate ion (HCO₃⁻) from the tissues to the lungs, where it is reconverted to CO₂ and expelled into the atmosphere as a body waste product.

Heme Break-down

- Heme breakdown occurs in macrophages of the reticuloendothelial system (liver, spleen, bone marrow).
- Biliverdin is converted to bilirubin by the enzyme, biliverdin reductase, using NADPH.
- High bilirubin levels cause jaundice, encephalopathy (kernicterus) and damage to multiple tissues.

- In jaundice urobilinogen (not stercobilinogen) accounts for the dark brown coloration of urine
- Conjugated bilirubin in the intestine is converted in to deconjugated bilirubin, then deconjugated bilirubin further converted to either urobilinogen (urine) or stercobilinogen (stool).

Jaundice

- Deposition of excess bilirubin in tissues causing yellow discoloration of the skin and sclera of the eyes.
- Hemolytic jaundice is due to excess red blood cell lysis and breakdown of heme to bilirubin. The liver's capacity to conjugate bilirubin is overwhelmed and unconjugated bilirubin accumulates in the blood.
- Obstructive jaundice occurs due to a failure of the biliary system to excrete conjugated bilirubin, for example due to gallstones or tumors in the bile duct.
- Hepatocellular jaundice is due to liver damage, for example from cirrhosis or hepatitis. This results in reduced conjugation of bilirubin and elevated bilirubin levels in the blood.
- Hemolytic jaundice results from increased bilirubin production; obstructive jaundice and hepatocellular jaundice result from defective conjugation/excretion of bilirubin.

Genetic causes of jaundice

Crigler-Najjar syndrome, Type 1-

- Due to inherited severe reduction or absence of UDP glucuronyl transferase. Associated with kernicterus (encephalopathy due to excess bilirubin). Severe jaundice (unconjugated bilirubin)

Crigler-Najjar syndrome, Type 2

- Reduced levels of UDP glucuronyl transferase, but less severe than Type 1 disease. Moderate non-hemolytic, largely unconjugated jaundice. Rarely involves kernicterus.

Gilbert's syndrome

- Mildly elevated unconjugated bilirubin levels which can cause mild jaundice. No treatment required- reassure the patient!

Newborn jaundice

- UDP glucuronyl transferase in the liver (hepatocytes) is low in the fetus and gradually increases after birth, reaching adult levels at about 4 weeks after birth.
- Premature babies lack enough UDP GUT to conjugate excess bilirubin.
- Phototherapy- convert's unconjugated bilirubin to soluble products that are easily excreted in bile without needing to be conjugated to glucuronic acid.

Biomarkers of blood proteins in clinical diagnosis

Albumin and globulin

- A biomarker has been defined as a characteristic that is objectively measured and evaluated as an indicator of normal biologic processes, pathogenic processes, or pharmacologic responses to a therapeutic intervention
- The plasma of patients with liver disease often shows a decrease in the ratio of albumin to globulins (decreased albumin-globulin ratio)
- The synthesis of albumin decreases relatively early in conditions of protein malnutrition, such as kwashiorkor

*COMPONENTS, CHARACTERISTICS AND
FUNCTION OF BLOOD*

- **Blood** is a specialized **connective tissue** that consists of **formed elements** (erythrocytes, leukocytes, and platelets) and a fluid component called **plasma**. The volume of blood in an average human adult is approximately 5 L (7-8% of body weight).
- Blood transports nutrients, waste products, hormones, proteins, ions, oxygen (O₂), carbon dioxide (CO₂), and formed elements.
- It also **regulates body temperature** and assists in **regulation of osmotic and acid-base balance, P.H 7.4**, and is **3X thicker and denser than water**.
- **Plasma** consists of **90% water**; **9% organic compounds** (such as proteins, amino acids, and hormones); and **1% inorganic salts, dissolved gases, and nutrients**.
- Main plasma proteins (about equally divided b/n Albumin and the different globulins)
 - **Albumin**; a small protein **preserves osmotic pressure** in the vascular system and helps **transport some metabolites**.
 - **Gamma-Globulins**; are **antibodies** (immunoglobulins). **Alpha-Globulins** and **beta-globulins**; transport metal ions (e.g., iron and copper) and lipids (in the form of lipoproteins).
 - **Fibrinogen**, (converted into fibrin during blood clotting)
 - **Complement proteins (C1-C9)** are part of the innate immune system,
- **Serum** is the yellowish fluid that remains after blood has clotted. It is similar to plasma, **but lacks fibrinogen and clotting factor**.
- **Hematocrit** (expressed as a %) is an estimation of the volume of packed erythrocytes per unit volume of blood. Normal values (40% to 50% in **adult men**), (35% to 45% in **adult women**).

- **Three** layers form after centrifugation (from the top- **Plasma-buffy coat(WBC's)-Hematocrit**)
- Important indices:
 - 1. **MCV: Mean Cell Volume** ($\text{Hct} \times 10 / \text{RBC No} = 87-92 \text{ FL}$) (If low=microcytic)
 - 2. **MCH: Mean Cell Hemoglobin** ($\text{Hg} \times 10 / \text{RBC No} = 30 \text{ Pg/cell}$) (is Hg conc. In a single cell) (low=hypochromic anemia)
 - 3. **MCHC: Mean Cell Hemoglobin Concentration** ($\text{Hg} / \text{Hct} = 333 \text{ g/l}$) (provides average Hg content in the mass of circulating RBC)
- **Hemostasis** involves arrest or stoppage of bleeding.
- Platelets are blood cells responsible for blood clotting. **3-A's of platelets:**
 - A) **Adhesiveness**: platelets stick when they come in contact with wet and rough surfaces (prevented from sticking to normal endothelium by a **glycoprotein in their membrane**)
 - B) **Aggregation**: When platelets are activated, they usually group together (b/c of **ADP and Thromboxane A₂** found in their cytoplasm).
 - C) **Agglutination**: Platelets clump together and form clots.
- 3-stages of Hemostasis:
 - 1) Vascular spasm of small blood vessels: involves myogenic contraction and the release of vasoactive substances
 - 2) Platelet plug: involves the aggregation of platelets that stick to injured vessels (facilitated by protein called **von Willebrand's** factor released from plasma and leaks into the traumatized tissues), after which platelets release ADP and Thromboxane-A₂ from their cytoplasm to attract more platelets. {prostacyclin and NO, from normal endoth, have vasodilatory and anti-aggregation functions}

- 3) Coagulation: This is a phase where blood loses its fluidity and becomes a jelly-solid mass. Occurs through either the **extrinsic** or **intrinsic** pathways. (Extrinsic pathway is faster)
 - **Extrinsic:** Traumatized tissue releases Thromboplastin-Tissue Thromboplastin converts VII to VIIa(active)- VIIa and Ca^{2+} convert factor X to its active form Xa - Factor Xa + Ca^{2+} VIIa produce Prothrombin activator -Prothrombin activator converts **prothrombin to thrombin which converts- Fibrinogen to thread like mesh called Fibrin monomer**(in bold is a pathway common to both intrinsic and extrinsic)
 - **Intrinsic:**mechanism begins with trauma to the blood itself or exposure of the blood to collagen at injured blood vessel wall.Blood then activates Factor XII to its active form XIIa .Then cascade of reactions continue until the formation of activated Factor Xa (Factor VIII + IX + platelet phospholipids (tissue thromboplastin) are involved in the activation).Activated Xa combines with Ca^{2+} and thrombin to produce the Prothrombin activator. This is followed by the common pathway.
- Clotting normally prevented by: **Smoothness of the endothelial** surface (prevents contact activation of intrinsic pathway), **Glycocalyx** on endotheliumsurface (repels clotting factors and platelets) or**Thrombomodulin** (binds thrombin and factors 5 and 8)
- **Clotting proteins** are made in the liver and require Vit K; hence factors affecting liver function or Vit k absorption will lead to bleeding disorders. (E.g. - Deficient fat absorption- decreased vit k absorption- No clotting enzymes)
- Prothrombin time (PT)-This test assesses the extrinsic and common coagulation pathways.

- Partial thromboplastin time (PTT): This test assesses the intrinsic and common clotting pathways.
- **Fibrinolysis:** The protein **plasminogen** is among the serum proteins that are adsorbed into the clot at the time of its formation. The cleavage of plasminogen by its activator (tissue plasminogen activator), produces the enzyme, **plasmin**, which **breaks fibrin** threads. The breakdown products of the fibrin mesh are released from the blood clot and are scavenged by macrophages.

*BLOOD CELLS GENERAL PATHOLOGY,
BLOOD TYPING AND CO POISONING*

- **Anemia-** decrease in red blood cell count, hemoglobin, or hematocrit (False Anemia can be caused by fluid retention-a drop in hematocrit).
- **Clinical signs and symptoms:** Patients appear pale (pallor), weakness, fatigability, dyspnea, hypoxia (causes fatty change in the liver, myocardium, and kidney, CNS hypoxia can cause headache, dimness of vision, and faintness)
- Anemia may be caused by two major mechanisms:
 - 1)Decreased red cell production resulting from:
 - Hematopoietic cell damage from infection, drugs, radiation, and other similar agents
 - Deficiency of factors necessary for heme synthesis (iron mostly and leads to hypochromic and microcytic anemia) or DNA synthesis (vitamin B12 or folate and leads to macrocytic anemia)
 - 2)Increased red cell loss due to: external blood loss or red blood cell destruction (hemolytic anemia)
- Anemias of blood loss-
 - A) Acute: volume lost-compensation from interstitium, decreased O₂-EPO secretion-Reticulocytosis (up to 15% in 7 Days){Cells are initially normal sized and colored(NCNC)}
 - B) Chronic- effects become apparent when loss exceeds replacement by marrow (Fe stores depleted), is initially NCNC, but after Fe depletion becomes hypochromic and microcytic.
- Hemolytic Anemias- have the following in common:
 - 1) Premature Destruction and shortened life span,
 - 2) Increased EPO and RBC production,

- 3) Accumulation of breakdown products.
- Hemolysis can be Extravascular (characterized by splenomegaly and jaundice, and is more common) or Intravascular (hemoglobinemia, hemoglobinuria, hemosiderinuria, Jaundice, and methemoglobin in urine making it brown); include
 - A) Hereditary Spherocytosis: defect in cell membrane (usually spectrin)-spheroid RBC's that are less deformable, sensitive to sequestration and destruction by spleen (cause of Splenomegaly) - shortened RBC life span (corrected by Splenectomy).
 - Dx- Family History, Jaundice and splenomegaly. (*Inherited in an autosomal dominant manner*).
 - Course is stable, but an aplastic crisis can be brought on by parvovirus infection.
 - B) Sickle cell Anemia (*homozygous for sickle cell gene*): sixth codon of β -globin, replacement of a glutamate residue with a valine residue, resultant Hemoglobin Hbs undergoes polymerization when deoxygenated and the RBC becomes sickle shaped (*repeated bouts make it irreversibly sickled*). Sickled RBC's obstruct microcirculation and are sensitive to hemolysis. There is extensive extra-medullary hematopoiesis which causes splenomegaly (*spleen becomes enlarged by red pulp congestion-splenic infarct, fibrosis leading to auto-splenectomy*), bone marrow is hyperplastic, repeated infarctions in the lungs and acute chest syndrome (*an aplastic crisis can be triggered by parvovirus*). Infants do not become symptomatic until they reach 5 or 6 months of age (*b/c HbF interferes with polymerization of Hbs*)

- Sickle cell trait: Hemoglobin S in the heterozygous form leads to sickle cell trait, which is generally without clinical consequence (*b/c HbA interferes with polymerization of Hbs*)
 - Electrophoresis is also used to demonstrate the presence of HbS and exclude other, such as HbSC diseases.
- C) Thalassemia-
 - 1)Beta Thalassemia- mutations in Gene coding Beta globulin,homo(Major)- or hetro(minor/trait)- zygoty for the trait(*Diminished or no synthesis*) - resembles Fe deficiency anemia(*microcytic and hypochromic*), alpha globulin accumulates in cell and causes reduced life span, marrow hyperplasia and extra-medullary hematopoiesis(*can cause Skeletal Abnormalities*), reduced hepcidin(*prevents excess Fe absorption*) and subsequent hemochromatosis cause CVS disease(*B-thalassemia minor is more common and patients are for the most part normal or mildly anemic*)
 - 2) Alpha thalassemia- inherited deletions that result in reduced or absent synthesis of α -globin chains. D/t forms-**Silent Carrier** (*1 α globulin gene deleted-no signs and symptoms*), **Thalassemia trait** (*2 α globulin genes deleted-no signs and symptoms*), **Hemoglobin H Disease** (*deletion of three α -globin genes-tetramers of β -globin, called HbH, which has high O₂ affinity and causes hypoxia*) **Hydrops Fetalis**(*deletion of all four α -globin genes. γ -globin chains form hemoglobin Barts, that have high affinity for oxygen and cause hypoxia, this form is lethal in utero without transfusion*)
 - D) Paroxysmal Nocturnal Hemoglobinuria- acquired mutations in PIGA gene- deficient in three GPI-linked proteins-improper complement activation at night(*b/c of ph drop*)- Intravascular RBC

hemolysis.(*RX with C5a complement inhibitor or cure with marrow transplant*).

- E) Immuno-hemolytic Anemia-d/t types
 - 1) Warm Ab (*most common form*)- IgG autoantibodies that react with red cell surface antigens. These antibodies are optimally active at 37°C, and are usually secondary to SLE or Hodgkin's and non-Hodgkin's lymphomas.
 - 2) Cold agglutinin disease is mediated by IgM antibodies optimally active at temperatures below 30°C (*cold antibodies, cold agglutinins*) (*acute form is secondary to mono- or M.pneumoniae infection*)
- Anemias of Diminished Erythropoiesis: most common are those caused by nutritional deficiencies, followed by those that arise secondary to renal failure (*No EPO*) and chronic inflammation.
 - A) Megaloblastic Anemias- 2 forms
 - 1) Pernicious anemia: autoimmune gastritis- no intrinsic factor- diminished vitamin B12 absorption-no DNA synthesis. Course is progressive without therapy.
 - Signs- low serum vitb12, hyper segmented granulocytes, leukopenia, increased homocysteine and methyl malonic acid in the serum. Upon parenteral admin of vitamin B12 marked reticulocytosis is observed in 5 days.
 - 2) Folate deficiency: no purine synthesis-no DNA synthesis(clinical signs of 1 and 2 are identical and serum levels of folate/vitb12 need to be checked)
 - B) Fe deficiency anemia: causes dietary lack, impaired absorption, increased requirement, and chronic blood loss. Produces a

hypochromic microcytic anemia. Fe reserves first depleted (*lowered serum Fe and transferrin saturation*) and then anemia occurs (*accompanied by low serum iron, ferritin, and transferrin saturation levels*).

- Signs- koilonychias, alopecia, and intestinal malabsorption, pica.
Rx if uncomplicated, oral iron supplementation produces an increase in reticulocytes in about 5 to 7 days.
- Anemia of Chronic Disease: Occurs in the setting of persistent systemic inflammation associated with low serum iron, reduced total iron-binding capacity, and abundant stored iron in tissue macrophages. IL-6-causes increased hepcidin production .RBC's can be NCNC or similar to Fe deficiency anemia. (*Only successful treatment of the underlying condition reliably corrects the anemia*)
- Aplastic Anemia: Aplastic anemia refers to chronic primary hematopoietic failure and attendant pancytopenia. In the majority of patients autoimmune mechanisms are suspected, but inherited or acquired abnormalities of stem cells also seem to contribute. In known etiology drugs/chemicals are the most common, radiation is a possible cause.
- Signs:Thrombocytopenia, Neutropenia, No Splenomegaly, Reticulocytopenia. (*Can be a pure Red Cell Aplasia in which only RBC progenitors are affected*)
- Myelophthisic anemia describes a form of marrow failure in which space-occupying lesions replace normal marrow elements.
- Polycythemia: denotes an abnormally high red cell count, usually with a corresponding increase in the hemoglobin level (*False Polycythemia can be caused by volume loss and*).
 - True polycythemia is primary when it results from an intrinsic abnormality of hematopoietic precursors. The most common cause of

primary polycythemia is polycythemia vera , a myeloproliferative disorder associated with mutations that lead to erythropoietin-independent growth of red cell progenitors, and a less common cause is familial mutations in the erythropoietin receptor that induce erythropoietin-independent receptor activation.

- Secondary polycythemia when the red cell progenitors are responding to increased levels of erythropoietin.

➤ **HEMOPHILIA**

- ✓ Hemophilia is a bleeding disease that occurs almost exclusively in males. In 85 per cent of cases, it is caused by an *abnormality or deficiency of Factor VIII*; this type of hemophilia is called *hemophilia A* or *classic hemophilia*.
- ✓ In the other 15 per cent of hemophilia patients, the bleeding tendency is caused by deficiency of Factor IX. Both of these factors are transmitted genetically by way of the female chromosome. Therefore, almost never will a woman have hemophilia because at least one of her two X chromosomes will have the appropriate genes. If one of her X chromosomes is deficient, she will be a *hemophilia carrier*, transmitting the disease to half of her male offspring.
- ✓ Depending on the character of genetic deficiency, the bleeding trait in hemophilia can have various degrees of severity. Bleeding usually does not occur except after trauma, but in some patients, the degree of trauma required to cause severe and prolonged bleeding may be so mild that it is hardly noticeable, for instance, bleeding can often last for days after extraction of a tooth.
- ✓ Factor VIII has two active components, a large and a smaller component. The smaller component is most important in the intrinsic pathway for clotting, and it is deficiency of this part of Factor VIII that causes classic hemophilia.
- ✓ When a person with classic hemophilia experiences severe prolonged bleeding, almost the only therapy that is truly effective is injection of purified Factor VIII. The cost of Factor VIII is high, and its availability

is limited because it can be gathered only from human blood and only in extremely small quantities.

➤ THE ABO BLOOD GROUP AND THE RH BLOOD GROUP

THE ABO SYSTEM

- ✓ There are several groups of red blood cell antigens, but the major group is known as the **ABO system**. In terms of the antigens present on the red blood cell surface, a person may be *type A* (with only A antigens), *type B* (with only B antigens), *type AB* (with both A and B antigens), or *type O* (with neither A nor B antigens).
- ✓ Each person's blood type—A, B, or O—denotes the antigens present on the red blood cell surface, which are the products of the genes that code for these antigens.
- ✓ A person who is type A, therefore, may have inherited the A gene from each parent (may have the genotype **I^AI^A**), or the A gene from one parent and the O gene from the other parent (and thus have the genotype **I^Ai**). Likewise, a person who is type B may have the genotype **I^BI^B** or **I^Bi**. It follows that a type O person inherited the O gene from each parent (has the genotype **ii**), whereas a type AB person inherited the A gene from one parent and the B gene from the other (there is no dominant-recessive relationship between A and B).
- ✓ The immune system exhibits tolerance to its own red blood cell antigens. People who are type A, for example, do not produce anti-A antibodies. Surprisingly, however, they do make antibodies against the B antigen and, conversely, people with blood type B make antibodies against the A antigen.

- ✓ People who are type **AB** develop tolerance to both of these antigens, and thus do not produce either anti-A or anti-B antibodies. Those who are type **O**, by contrast, do not develop tolerance to either antigen; therefore, they have both anti-A and anti-B antibodies in their plasma.
- ✓ Before transfusions are performed, a *major cross match* is made by mixing serum from the recipient with blood cells from the donor. If the types do not match—if the donor is type **A**, for example, and the recipient is type **B**—the recipient's antibodies attach to the donor's red blood cells and form bridges that cause the cells to clump together, or **agglutinate**.
- ✓ Because of this agglutination reaction, the **A** and **B** antigens are sometimes called *agglutinogens*, and the antibodies against them are called *agglutinins*.
- ✓ Transfusion errors that result in such agglutination can lead to blockage of small blood vessels and cause hemolysis (rupture of red blood cells), which may damage the kidneys and other organs.
- ✓ Since type **O** red blood cells lack **A** and **B** antigens, the recipient's antibodies cannot cause agglutination of the donor red blood cells. Type **O** is, therefore, a *universal donor*—but only as long as the volume of plasma donated is small, since plasma from a type **O** person would agglutinate type **A**, type **B**, and type **AB** red blood cells. Likewise, type **AB** people are *universal recipients* because they lack anti-A and anti-B antibodies, and thus cannot agglutinate donor red blood cells. (Donor plasma could agglutinate recipient red blood cells if the transfusion volume were too large.)

Rh FACTOR

- ✓ This Rh antigen is termed D, and is often indicated as Rho (D). If this Rh antigen is present on a person's red blood cells, the person is **Rh positive**; if it is absent, the person is **Rh negative**. The Rh-positive condition is by far the more common.
- ✓ The Rh factor is of particular significance when Rh-negative mothers give birth to Rh-positive babies. Since the fetal and maternal blood are normally kept separate across the placenta, the Rh-negative mother is not usually exposed to the Rh antigen of the fetus during the pregnancy. At the time of birth, however, a variable degree of exposure may occur, and the mother's immune system may become sensitized and produce antibodies against the Rh antigen.
- ✓ If the woman does produce antibodies against the Rh factor, these antibodies could cross the placenta in subsequent pregnancies and cause hemolysis of the Rh-positive red blood cells of the fetus. Therefore, the baby could be born anemic, with a condition called *erythroblastosis fetalis*, or *hemolytic disease of the newborn*.
- ✓ Erythroblastosis fetalis can be prevented by injecting the Rh-negative mother with an antibody preparation against the Rh factor within 72 hours after the birth of each Rh-positive baby. This is a type of passive immunization in which the injected antibodies inactivate the Rh antigens and thus prevent the mother from becoming actively immunized to them.

➤ **HEMOLYTIC DISEASES AND HEMOGLOBINOPATHIES**

The most common hemolytic diseases and hemoglobinopathies are the following;

• **HEMOLYTIC DISEASES**

- ❖ **Erythroblastosis fetalis-** This disorder is a hemolytic disease of the fetus caused by maternal antibodies against fetal red cell antigens such as Rh antigens and ABO blood group antigens. The maternal antibodies cause hemolysis of the fetal red blood cells and fetal anemia.
- ❖ **Hemolytic Anemias-** The hemolytic anemias encompass processes that result in red blood cell destruction (hemolysis) after release from the bone marrow.

A common compensatory change that accompanies hemolytic anemias is an increase in production of red blood cell precursors, manifested by polychromasia of red blood cells and increased reticulocyte count.

Hemolytic anemias are classified according to site of red blood cell destruction. Extravascular hemolysis occurs in the spleen and liver by cells of the monocyte/macrophage system. Intravascular hemolysis involves the destruction of cells within the circulation.

• **HEMOGLOBINOPATHIES**

The hemoglobinopathies are caused by point mutations of the beta globin chain gene.

- **Sickle Cell Disease-** Sickle cell disease is caused by a point mutation resulting in a substitution of a valine for the normal glutamic acid in the sixth amino acid position in the beta globin chain gene (hemoglobin S).

- This substitution leads to a polymerization of the beta globin chain within the cytoplasm of red blood cells during deoxygenation, resulting in characteristics sickling of the cells.
 - Sickled red blood cells are less deformable and may lead to obstruction of the microcirculation with resultant tissue hypoxia and ischemic injury.
 - Sickle cell disease demonstrates highest incidence in persons of African ancestry and heterozygosity for hemoglobin S may offer some protection against infection by malaria.
 - Patients with sickle cell disease have severe normocytic or macrocytic anemia.
 - Patients who are homozygous for hemoglobin S (hemoglobin SS) demonstrate the full manifestation of sickle cell disease.
 - During the first several months of life, patients with sickle cell disease are asymptomatic because of the high levels of hemoglobin F present in circulation. However, overtime, patients primarily have circulating hemoglobin SS and develop an adaptive response to lifelong hemolysis.
- ❖ **Sickle cell Trait- Heterozygosity** for the hemoglobin S mutation is referred to as sickle cell trait. Patients with sickle cell trait are asymptomatic, except for instances of extreme oxidative stress such as deep sea diving.

In these patients the presence of hemoglobin A prevents polymerization of hemoglobin S and therefore sickling.
- ❖ **Hemoglobin C Disease- Hemoglobin C** disease results from a homozygous inheritance of hemoglobin that contains a substitution of a lysine for the normal glutamic acid molecule at the sixth amino acid position of the beta globin gene.

- Hemoglobin C precipitates within red blood cells, making them less deformable and therefore susceptible to removal by spleen. Patients may experience a mild chronic hemolytic anemia and splenomegaly.
- ❖ **Hemoglobin E Disease- Hemoglobin E** disease results from a homozygous inheritance of a hemoglobin that contains a substitution of a lysine for the normal glutamic acid at position 26 of the beta globin gene. This mutation occurs at a splice site, which leads in addition to decreased gene transcription and an unstable messenger RNA.
 - ❖ Patients with homozygous EE have a mild microcytic anemia, decreased MCV, erythrocytosis, and target cells. Greater than 90% of the hemoglobin is made up of hemoglobin E.

➤ **Carbon monoxide (CO) Poisoning**

- ✓ CO is an odorless gas that results from the incomplete combustion of organic substances. Environmental CO is derived mainly from automobile exhaust emissions, fires, and in some cases, home heating systems.
- ✓ Chronic poisoning by CO develops because carboxyhemoglobin, once formed, is remarkably stable. As a result, with low-level persistent exposure to CO, carboxyhemoglobin may accumulate to a life-threatening concentration in the blood.
- ✓ The slowly developing hypoxia can insidiously evoke widespread ischemic changes in the brain; these changes are particularly marked in the basal ganglia and lenticular nuclei.
- ✓ With cessation of exposure to CO, the patient usually recovers, but there may be permanent neurologic damage.
- ✓ Acute poisoning by CO generally is a consequence of accidental exposure or suicide attempt. In light-skinned people, it is marked by a characteristic

generalized cherry-red color of the skin and mucous membranes, color imparted by carboxyhemoglobin.

- ✓ If death occurs rapidly, morphologic changes may not be present; with longer survival, the brain may be slightly edematous and exhibit punctate hemorrhages and hypoxia-induced neuronal changes. These changes are not specific; they simply imply systemic hypoxia.
- ✓ In victims who survive CO poisoning, complete recovery is possible; however, sometimes impairments of memory, vision, hearing, and speech may remain

BLEEDING DISORDERS
(HAEMMORHAGIC DIATHESIS)

- **Bleeding Disorders: (Hemorrhagic Diathesis)** Excessive bleeding can result from:
 - Increased fragility of vessels,
 - Platelet deficiency (thrombocytopenia) or dysfunction, and
 - Derangement of coagulation, alone or in combination
 - **Bleeding Disorders Caused by Vessel Wall Abnormalities** (non-thrombocytopenic purpuras):
- Causes: *microbial damage to the microvasculature* (vasculitis) and **DIC**, *Drug reactions* (drug-induced immune complexes in vessel walls), *Scurvy and the Ehlers-Danlos Syndrome* (collagen disorders), *hereditary hemorrhagic telangiectasia* (dilated, convoluted blood vessels with thin walls that bleed readily) or Perivascular amyloidosis. **Thrombocytopenia** (below 100,000 platelets/ μ L): 20,000 to 50,000 platelets/ μ L can aggravate post-traumatic bleeding. Spontaneous bleeding occurs below 20,000 platelets/ μ L.
- Thrombocytopenia is associated with a normal PT and PTT.
 - Causes of thrombocytopenia: decreased platelet production, decreased platelet survival, sequestration, dilution.
- **Chronic Immune Thrombocytopenic Purpura (ITP)**: caused by autoimmune Ab's (usually IgG) directed against platelets (and sometimes against megakaryocytes).
- CITP is primary (idiopathic) if no known cause is found, or secondary to another condition (HIV infection, B-cell neoplasm).
- Most common in adult women typically under 40 years of age. Manifests with melena, hematuria, or excessive menstrual flow. Rx with glucocorticoids, splenectomy anti-CD20 Ab's.

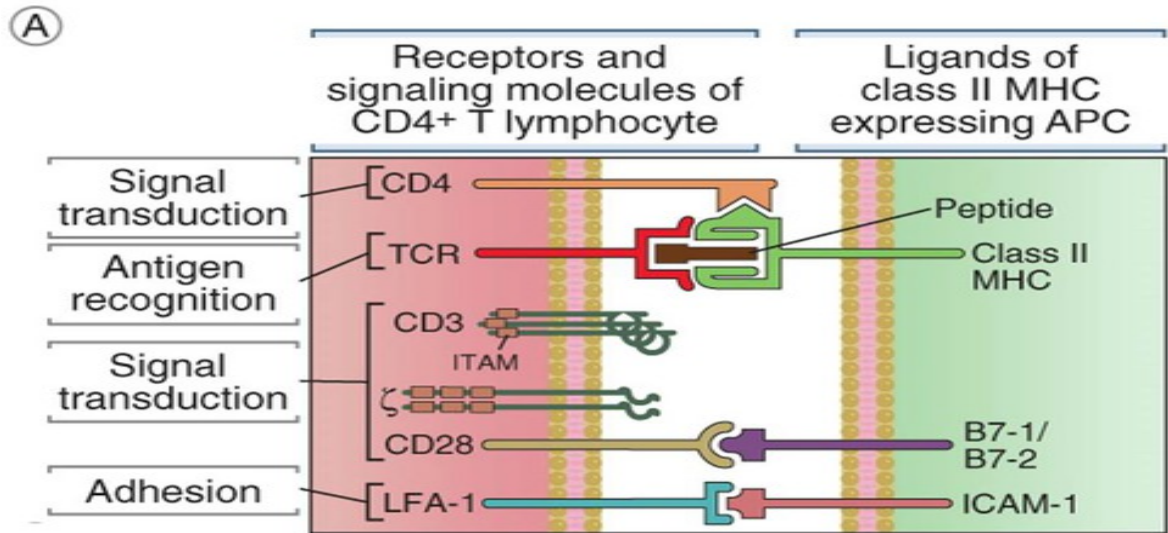
- *Acute Immune Thrombocytopenic Purpura*: also caused by auto-AB, but is primarily seen in children following a viral infection and is self-limited and usually doesn't require Rx.
- *Drug-Induced Thrombocytopenia*: drugs bind platelet glycoproteins and create antigenic determinants that are recognized by antibodies-platelet destruction
- *HIV-Associated Thrombocytopenia*: CD4 and CXCR4 are found on megakaryocytes-HIV-infected megakaryocytes are prone to apoptosis and their ability to produce platelets is impaired.
 - **Bleeding disorders related to defective platelet functions**: can be inherited or acquired.
- Inherited disorders of platelet function can be defects in adhesion, aggregation, or platelet secretion.
- *Abnormalities in Clotting Factors*: inherited ones usually affect 1 clotting factor, as opposed to acquired defects which usually affects multiple clotting factors (e.g.-decreased protein synthesis or decreased half-lives).
- Liver parenchymal disease prevents many clotting factors from being made.
- Manifest as large post-traumatic ecchymosis or hematomas, or prolonged bleeding after any surgical procedure.
- The most common inherited deficiencies affect factor VIII (hemophilia A), and factor IX (hemophilia B).
- Hemophilia A is the most common hereditary disease associated with life-threatening bleeding. It is caused by mutations in factor VIII, which is an essential cofactor for factor IX in the coagulation cascade.

- Hemophilia A is inherited as an X-linked recessive trait and thus affects mainly males and homozygous females.
- The most severe deficiencies result from an inversion involving the X chromosome that completely abolishes the synthesis of factor VIII.
- Patients have a prolonged PTT and a normal PT.
- Hemophilia B inherited as an X-linked recessive trait and shows variable clinical severity. Patients have PTT is prolonged and the PT is normal.
- The above 2 hemophilia's are clinically indistinguishable, and a factor assay is required to tell them apart.
 - Signs: tendency toward easy bruising and massive hemorrhage after trauma or operative procedures (Petechiae are absent)
- *Disseminated Intravascular Coagulation (DIC)*: chronic thrombo-hemorrhagic disorder characterized by the excessive activation of coagulation, which leads to the formation of thrombi in the microvasculature of the body(*that lead to occlusion and infarcts*) and uses up clotting factors, fibrin and platelets(*hemorrhage due to depletion of factors required for hemostasis and the activation of fibrinolytic mechanisms; or both*).
- Caused by pathologic activation of either clotting mechanisms or non-functional clot inhibiting system.
- Two major mechanisms trigger DIC:
 - Release of tissue factor or thromboplastic substances into the circulation (can come from neoplastic cells(acute promyelocytic leukemia cells),placenta in obstetric complications)
 - Widespread injury to the endothelial cells

*CMI, ANTIGEN RECOGNITION, HUMORAL
IMMUNITY AND ITS EFFECTOR MECHANISM*

Cell-mediated immunity

- Phases of T cells is responses of T lymphocytes to cell-associated microbial antigens involving a series of sequential steps that result in an increase in the number of antigen-specific T cells and the conversion of naïve T cells to effector cells includes
 - **Clonal proliferation** of antigen-specific T cells resulting in a rapid increase in the number of antigen-specific lymphocytes
 - **Differentiation** results in conversion of **naïve T cells**, whose function is to recognize microbial antigens, into **effector T cells**, whose is to eliminate microbes
 - **Memory T cells** progeny of T cells that have proliferated in response to antigen which are long-lived and functionally inactive and circulate for months or years, ready to rapidly response to repeat exposures to the same microbe (both 4 and 8)
- Activation of T cell response requires multiple receptors
 - Recognition of ligands on APCs
 - TCR=MHC associated peptides
 - CD4 or CD8 co-receptors = MHC molecules
 - Adhesion molecules = strengthen the binding of T cells to APCs
 - Receptors for co-stimulators recognize = second signals provided by the APCs



- TCR recognition of the MHC molecules alone doesn't transmit signal transduction. Two or more TCRs need to come together to transmit the signals.
- TCR is covalently associated with a complex of the 3 proteins that make up CD3 and with a homodimer of another signaling protein called the ζ chain
- There are also other co-stimulators involved in these processes like CD28 that reacts with B7 which co-stimulates but when CTLA 4 binds with B7 then there is negative regulation
- Other molecules like ICAM -LFA 1 (APCs) and VCAM-VLA 4 (endothelium) facilitate adhesion and signal transduction
- Poly clonal activation can be achieved by super antigens or Abs specific for the TCR associated CD3 proteins meaning that these molecules play a great role
- Adhesion molecules play a great role in that they stabilize the binding long enough that the necessary signal threshold is achieved
- Biochemical pathways of T cell activation involve expression of proteins that are involved in proliferation, differentiation, and effector functions. Their response involves recruiting enzymes, adaptor proteins, and transcription factors
 - cFOS -transcription factor

- IL2-growth factor induce proliferation
 - CD40-effector molecule
 - CD69 -migration factor
- After activation T cells begins secretion of cytokines

General properties of cytokines

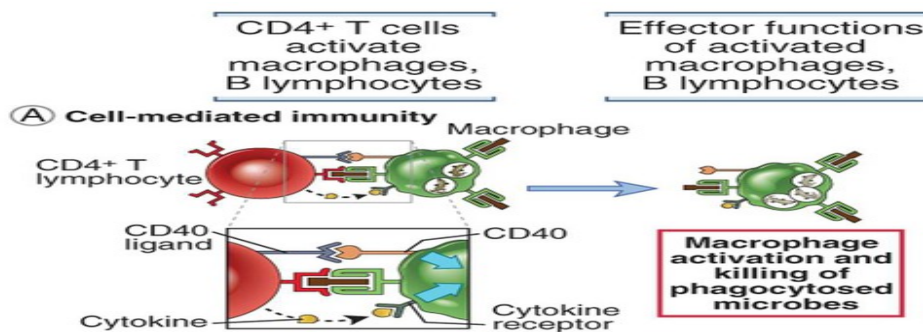
A General properties of T cell cytokines	
Property	Significance
Produced transiently in response to antigen	Provides cytokine when needed
Usually acts on same cell that produces the cytokine (autocrine) or nearby cells (paracrine)	Systemic effects of cytokines usually reflect severe infections or autoimmunity
Pleiotropism: each cytokine has multiple biologic actions	Provides diversity of actions but may limit clinical utility of cytokines because of unwanted effects
Redundancy: multiple cytokines may share the same or similar biologic activities	Blocking any one cytokine may not achieve a desired effect

B Biologic actions of select T cell cytokines		
Cytokine	Principal action	Cellular source(s)
Interleukin-2 (IL-2)	Survival, proliferation, and differentiation of effector and regulatory T cells	CD4+ and CD8+ T cells
IL-4	B cell switching to IgE	CD4+ T cells, mast cells
IL-5	Activation of eosinophils	CD4+ T cells, mast cells
Interferon- γ (IFN- γ)	Activation of macrophages	CD4+ and CD8+ T cells, natural killer cells
IL-17	Stimulation of acute inflammation	CD4+ T cells; other cells
TGF- β	Inhibition of T cell activation; differentiation of regulatory T cells	CD4+ T cells; many other cell types

- 1 or 2 days after activation, T cells begin **clonal proliferation**. CD8+ T cells, 1 in 10^5 or $10^6 \rightarrow 10\%$ to 20% of all lymphocytes in the body [100,000 fold]. CD8 T cells proliferate in a much more larger scale than CD4 T cells by **1000**

fold but this proliferation doesn't include the bystander T cells that don't recognize the antigen

- Concurrently with expansion some progeny of T cells differentiate to its effector cell that leave the lymphoid tissue and migrate to the site of infection .this process is result of changes in genes encoding cytokines in CD4 T cells and cyto-toxic proteins in CD8 T cells
- CD4 T cells differentiate by producing CD40 Ligand .the receptor of this ligand is present in macrophages in B lymphocytes and dendritic cells. Provides positive feedback mechanism or amplification for APC mediated activation of T cells



CD4 T cells sub population

- TH1 cells stimulate phagocyte-mediated ingestion and killing of microbes, a key component of cell-mediated immunity. These cells release IFN gamma. This acts on B cells and macrophages and enhances opsonization and enhanced killing
- TH2 cells stimulate phagocyte-independent, eosinophil-mediated immunity, which is especially effective against helminthic parasites .these cells release IL4 which stimulate release of IgE antibodies

- TH17 cells secrete the cytokines IL-17 and IL-22 and are the principal mediators of inflammation in a number of immunologic reactions
- There can also be a regulatory T cell
- As stated above, some progeny of the activated T-cells differentiate in to memory T cells. They prevail even after infection has been eradicated but require signals delivered by certain cytokines (IL-7) to stay alive. Types:
 - Central memory-lymphoid organs, responsible for rapid clonal expansion after re-exposure to antigen
 - Effector memory-localize in mucosal tissue & mediate rapid effector functions

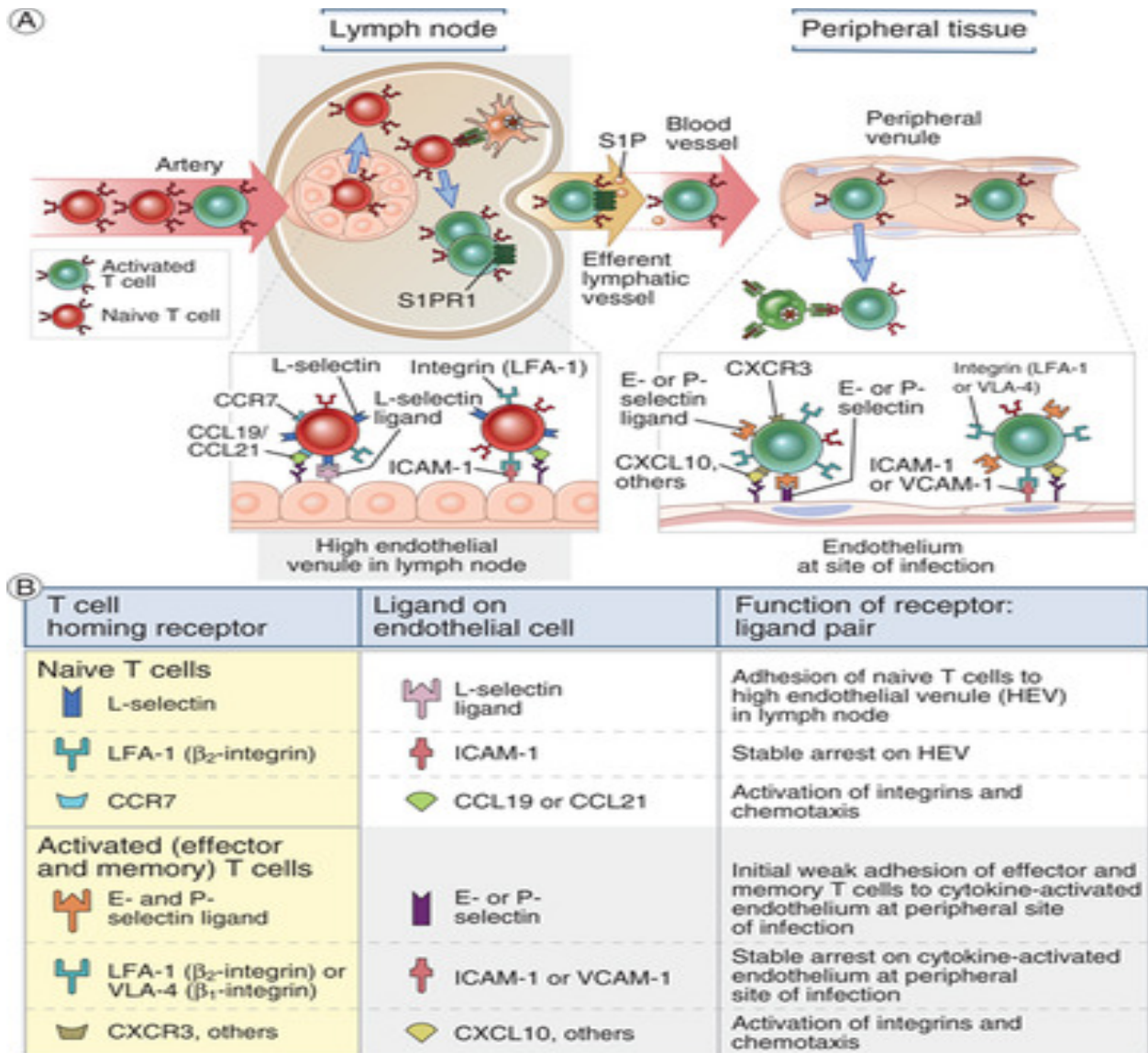
BARRIERS TO T CELL IMMUNE RESPONSE

- Naive T cells have to find the antigen
 - APCs
- correct type of T cells must respond for antigens from extracellular compartment and intracellular compartment
 - Selectivity is determined by the specificity of the CD4 or CD8 co-receptors for MHC-II and I
- T cells must interact with antigen-bearing APCs long enough to be activated
 - Adhesion molecules
- T cells should respond to microbial antigens but not to harmless proteins
 - Preference of microbes is maintained because T cell activation requires costimulators that are induced on APCs by microbes

- Antigen recognition by a small number of T cells must lead to a response that is large enough to be effective By several amplification mechanisms induced by microbes and activated T cells themselves lead to enhanced T cell activation

Effector response

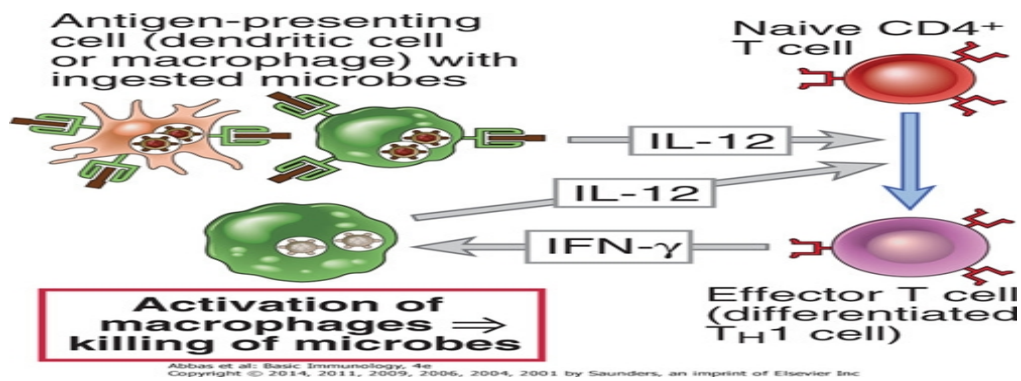
- CD4+ T cells activate phagocytes to destroy microbes residing in the vesicles of these phagocytes
- CD8+ CTL kill any cell containing microbes or microbial proteins in the cytoplasm
- Migration of the cells to the infection site is mediated by E-selectin ,P-selectin and integrins different chemokines secreted at the inflammatory site



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- IFN gamma released by activated CD4 T specifically to TH1 subpopulations cells upon stimulation by IL12 released by the macrophages activate macrophages. So it is a kind of a reciprocal effect



- Several cytokines produced by TH2 cells, including IL-4, IL-10, and IL-13, inhibit the microbicidal activities of macrophages. IL-4 and IL-13 also can activate macrophages to express mannose receptors and IL-13 acts on fibroblasts to increase collagen synthesis and fibrosis. We call this the alternative macrophage activation.
 - For example if we get infected by mycobacterium leprae a bacteria which lives inside macrophages the type of T cells that on them determine the outcome; if TH1 cells act on them then the macrophages will kill these pathogens but if TH2 cells act on them they will disseminate and cause the disease so which T cell subtype acts on the macrophage will determine the disease outcome.

Antigen presentation

Terms:

- **Immunogen** is any agent capable of inducing an adaptive immune response
- **Antigen** is any agent capable of being recognized by the adaptive immune system
- **Immunogenicity** is the ability of an antigen to stimulate an immune response
- **Hapten** - A substance that is non-immunogenic but which can react with products of a specific immune response
- **epitope or antigenic determinant** structural portion of antigen that bind to antibodies or TCR
- **Antibody (Ab)** specific protein which is produced in response to an immunogen and which reacts with an antigen
 - size of an epitope for Ab binding is about 5-7 aa residue
- antigen-binding site region of Ab where an epitope is binding
 - size of an epitope that binds corresponding TCR is about 10-15 aa residues
- **Anchor residues:** aa residues of antigenic peptide binding MHC molecules
- Polysaccharides antigen may have single epitope or multiple epitopes of same specificity.
- Protein antigen have a combination of different epitopes on same molecule.

Factors affecting immunogenicity

- nature of the immunogen
- type of antigen
 - T-independent antigen
 - Excite B cells without the help of T cells(polysaccharides)
 - **Polymeric structure**
 - same antigenic determinant repeated many times
 - **Polyclonal activation of B cells**
 - activate B cell clones specific for other antigens
 - Type 1 (polyclonal activators) and Type 2(not)
 - **Resistance to degradation**
 - persist for longer periods of time and continue to stimulate
 - T-dependent antigens
 - Do not excite B cells directly they need the help of T cells
(proteins)

Antigenic determinants recognized by B cells (Abs)

- Compositions
 - Proteins, polysaccharides , nucleic acids,
 - Sequence (linear)or conformation (shape)
- Size
 - 4-8 residues
- Number
 - Limited number;Dominant epitopes found on the outer surface of the antigen

Antigenic determinant recognized by T cells

- Composition
 - Proteins sometimes lipids
- Size
 - 8-15
- Number
 - Limited to those that can bind to MHCs

Super antigens

- are antigens that polyclonal activate a large fraction of T cells (up to 25%) compared to 0001-.001% for normal antigens
- these usually are toxins but can also be viruses and other microorganisms

How do T cells recognize MHCs and antigens?

- T cells recognize
 - Peptide residues of antigens and also MHCs and APCs
- Different clones of T cells recognize antigens only if they are displayed by MHCs ;this is what we call **MHCs restriction**
- When microbes get in to our system they are recognized and captured by APCs and taken to the **secondary lymphoid organs** to be displayed for recognition by T cell
- Antigen processing: is breaking down of the protein molecule in to peptide so they can be displayed by MHC molecules
 - Exogenous antigens: are endocytosed and processed in the endocytic pathway.

- Endogenous antigens: are derived from antigens that have been processed in the cells cytoplasm.

➤ Different T cells bind to two distinct MHC molecules

- **CD4 T** cells bind to **MHC class II**: these cells presenting antigens are the **only ones called APCs**, and they display **extracellular** proteins internalized by the APCs .incorporated with the molecules in the **endocytotic vesicles**.
- **CD8 T** cells bind to **MHC class I**: the cells displaying this class of the molecule are called **target cells** and they display any protein processed in cytosol of any nucleated cells. Incorporated with the molecule in the **RER**.
- **Cross representation** :although not clear come exogenous antigens can gain access to MHC I molecules ,this maybe by rapture of their endosomal membrane or some have developed a retrograde pathway that takes from the endosome
- CD4 and CD8 are called **co receptors**; CD4 binds to the outer surface of the $\beta 2$ domain of the MHC II, CD8 binds to the $\alpha 3$.
- The expression of MHC class II molecules is induced and increased by IFN gamma for macrophages and dendritic cells respectively while IL4 increases the expression on B lymphocyte

- All nucleated cells can express MHC class I
- An MHC molecule can bind one peptide at a time but it has the ability to bind and express many different kinds of peptide (only peptides)
 - Peptides incorporated during MHC biosynthesis, assembly and transport
 - Only Peptide bound MHCs are stably expressed have very long offset so the T cells can locate them

Antigen recognition

- B and T lymphocytes differ in the types of antigens they recognize.
- The antigen receptors of B lymphocytes— namely, membrane-bound antibodies—can recognize a wide variety of macromolecules (proteins, polysaccharides, lipids, and nucleic acids), as well as small chemicals in soluble or cell surface-associated form. Therefore, B cell-mediated humoral immune responses may be generated against many types of microbial cell wall and soluble antigens.
- Most T lymphocytes, on the other hand, can see only peptide fragments of protein antigens, and can do so only when these peptides are presented by specialized peptide display molecules on host cells peptides meaning that they recognize peptide fragments presented by the major histocompatibility complex (MHC) displayed by antigen presenting cells (APCs dendritic cells) .
- They are similar in structure and are clonal they transmit biological signal

➤ BCR

- consist of regions/domains involved in
 - specific antigen recognition
 - structural integrity
 - effector functions
 - variable (V) and constant (C)
- what you have to remember is that for the receptors to recognize their ligands they need to bind so what is crucial here is that this binding is **non covalent it can be hydrogen bond or charged interaction.**
- main function of these receptors is to first **recognize** their specific ligand and **transmit the signals** needed for the function
 - transduction is mediated by a variety of polypeptides and this combination of the receptors and the signaling molecules is called a complex (**BCR and TCR complex**)
 - the portion of the antigen recognized are called **epitopes** or determinants
- when a ligand binds adjacent receptors are pulled together into an aggregate
 - cross linking so that signaling proteins can come to close proximity and facilitates phosphorylation by enzymes
- BCR -membrane antibodies (Ig)
 - composed of **four** polypeptide chains:
 - two identical **heavy** :made up of 1 V and 3-4 C domains (1V:3C)
 - **$\alpha, \delta, \epsilon, \gamma$, and μ : IgA, IgD, IgE, IgG, and IgM**
 - two identical **light** :made of 1 V and 1 C domains(1V:3C)
 - **lambda (λ) and kappa (κ)** in mammals
 - four chains form **Y-shaped** molecule
 - **disulfide bonds** attach

- each light chain to one heavy chain
- two heavy chains to each other
- **complementarity determining regions(CDRs)**
 - each variable regions of V_H or of V_L contains 3 hypervariable regions, or **CDRs** (1,2 & 3) CDR3
 - with greatest variability
 - form clefts to accommodate small molecules or flat surfaces to accommodate big molecules
 - located in junction of V and C regions
 - contributes to most of antigen binding
- **Fab (fragment antigen-binding) region**
 - contains whole light chain (with its single V and C domains) attached to V and first C of heavy chain
 - bind antigen
- **Fc region (fragment crystalline)**
 - remaining heavy chain C domains
 - responsible for biologic activity & **effector functions**
- **Hinge region**
 - between Fab and Fc
 - flexible portion
 - allow two Fab regions to move independently

Antigen antibody reaction

- **Affinity**
 - strength with which one antigen-binding surface of an antibody binds to one epitope of an antigen
 - is expressed as dissociation constant(K_d)

- molar **concentration of an antigen** required to occupy **half the available antibody molecules** in solution
- Affinity maturation
 - repeated stimulation increases affinity
- Avidity
 - accumulated strength of multiple affinities of individual binding interactions (**functional avidity**)
 - difference with affinity :affinity is a strength of a single interaction
- Cross reaction
 - binding of antibodies produced against one antigen to other **similar epitopes**
- TCR
 - recognizes peptide antigens displayed by MHC
 - a membrane-bound heterodimer composed of an α chain and β chain (1V and 1C each)
 - 5-10% of T cells in the body express receptors of γ and δ
 - recognize a variety of protein & non-protein
 - Not displayed by classical MHC molecules
 - Abundant at epithelial surfaces
 - NKT cells
 - express $\alpha\beta$ TCRs
 - recognize lipid antigens displayed by CD1d
- clonal selection theory
 - Each lymphocyte bears a single type of receptor with a unique specificity (by V(D)J) recombination

- The differentiated effector cells derived from an activated lymphocyte will bear receptors of identical specificity as the parental cell
- Those lymphocytes bearing receptors for self-molecules will be deleted at an early stage

➤ Maturation of lymphocytes

- Proliferation of immature cells
- Expression of antigen receptor genes
- Selection of lymphocytes that express useful antigen receptor
- Then exposure to antigen
 - Negative selection if a lymphocyte has a strong recognition for a specific antigen then it is deleted
 - Positive selection if a lymphocyte has a weak recognition for an antigen then it is selected to mature

HUMORAL IMMUNITY AND EFFECTOR RESPONSE

- mediated by antibodies
- Arm of adaptive immune response
- functions to neutralize and eliminate extracellular microbes and microbial toxins
- more important than cellular immunity
 - in defending against microbes with capsule rich polysaccharides and lipids, and against polysaccharide and lipid toxins
- B cells respond to, and produce antibodies specific for, many types of molecules, but T cell, the mediators of cellular immunity, recognize and respond only to protein antigens
- Antibodies are produced by B cells but only if they recognize antigen and differentiate to plasma cells which secrete antibody
- Naïve B cells have two membrane bound antibodies IgM and IgD which act as a receptor these receptors recognize antigens and activate the B cell which leads to clonal proliferation and differentiation to plasma cells
- one activated B cell may generate **4000 plasma cells** and can produce 10^{12} antibody molecules/day and the antibody produced have the same specificity as the parent cell receptor but in some cases the secreted antibodies may have different type of heavy chain this is called *heavy chain class switching*
- *Affinity maturation* is when there is a repeated exposure to a protein antigen then this increases the affinity of the antibody. *only on TH dependent activation*

- There are T dependent (proteins) and T independent (polysaccharides, lipids) Ab (antibody) responses based on the requirement of T cell help.
 - T helper cells play an important role in B cell activation in that they induce isotype switching and that they illicit a strong response; protein antigens would have a weak or low Ab response without TH cells. The Abs produced by the T dependent pathway have very high affinity and also good isotype switching

Subset of B cells

- Follicular B cells :reside in the follicles of the lymphoid tissue and make the bulk of T dependent responses.it gives long lived plasma cells
- Marginal zone B cells reside in the white pulp of the spleen and respond to blood born polysaccharides in the mucosa and peritoneum
- B-1 B cells respond to non-protein antigens in the mucosa
 - ❖ Marginal and B-1 B cells have receptors of limited diversity mainly IgM

Primary and Secondary responses

- Antibodies produced when first exposed to an antigen are **less in number** and **also low in affinity maturation** compared to the response produced when exposure in the second time .
- There is a lag in the eliciting of response in primary exposure usually 5-10 days but secondary response takes 1-3 days
- There is also high isotype switching in the secondary response
- Upon binding of an antigen the membrane Igs (mIg) gets cross linked together or cross-linked then this will initiate the needed signal and with the receptor associated proteins transducer signals .so this means that the receptors recognize the antigen but do not transmit the signals

- The receptors have ITAM s and these get phosphorylated and then the phosphotyrosine become a docking site for adapter protein
- Complement proteins provide second signals for B cell activation and this parallels with co-stimulator signal for T cells
- Complements get fragmented and coat the microbes and also bind to CR2 on B cells same time.
- Engagement of CR2 greatly enhances antigen-dependent activation of B cell responses.
- So function of activated B cell is the proliferation of, clonal expansion and differentiation and increased expression of cytokine receptors to interact with T helper cells in the lymphoid organs.

How do the TH cells specific for that antigen and the B cells find each other?

- TH cells reduce expression of chemokine receptor CCR7 (for T cell zones)and increase expression of CXCR5 which promotes migration to B cell follicles
- B cells however undergo the exact opposite changes so they would meet at the edge of the follicles
- B cells bind **conformational** epitope phagocytose process antigens and display them in MHC class II molecules
 - B cells and T cells recognize different epitopes of the same protein
- TH cells also activate B cells by expressing CD40L and cytokines just like when they activate macrophages
- Isotype switching(difference in constant region) is important for different antibodies to carry out different functions for example opsonization requires IgG1 and 3 bind to FC receptors in bacteria or viruses specific for the gamma

chain or for helminthes the IgE s recruit eosinophils by binding to the FC receptor specific for that chain ;this is highly influenced by cytokines

- IgE -helminthes by IL4 ,IgG -phagocytic response by IFN-gamma ,IgA - mucosal immunity
- Affinity maturation is due to point mutation in the V region

Antibody response to TH independent activation of B cells

- These antigens meaning polysaccharides and lipids have the same kind or peptide repeated many times so it's able to cross like many receptors together to initiate a signal .When there is extensive cross linking, then there is differentiation and expansion without TH cells.
- Regulation would be ;
 - gradual loss of B cells and also antibody feedback meaning antibody bound to antigen inhibit further antibody production when the antigen antibody complex binds to FcγRII inhibitory receptor on B cells

EFFECTOR MECHANISM OF HUMORAL IMMUNITY

- It is mediated by antibodies against extra cellular antigens and selective stages of intracellular antigens that must exist extracellularly and their toxins and
- function requires cooperation of phagocytes and complement components of innate immunity
- Abs produced by the B cells or plasma cells act on sites that are far away from their production –systemic
- After antigen exposure long live plasma cells continue to produce antibodies for immediate protection .B memory cells will increase production of antibodies when there is reexposure since they will become reactivated
- Activation of antibodies is when antigen binds antibodies so this insures that they exert their effector function only when they are needed (when they bind antigen)
- The heavy chain of the Abs determine their functions .the microbes causes class switching (by activating T cells to produce cytokines and express the CD40L) so that the appropriate Ab is secreted for the appropriate pathogen
- The Ab exert there effect through binding to the FC receptor on leukocytes that has the binding region and signaling region
 - IgM : classical complement activation and antigen receptor
 - IgG: opsonization ,classical complement pathway activation ,for feedback inhibition of B cell activation ,antibody dependent cell mediated cytotoxicity (macrophages),neonatal immunity
 - IgA: mucosal immunity
 - IgE :antibody dependent cell mediated cytotoxicity (eosinophills)and mast cell degranulation (immediate hypersensitivity reaction)
 - IgD : antigen receptor

Neutralization of microbes or their toxins

- Mediated by any isotype but mostly IgG is the most neutralizing isotype found in the blood

Antibody mediated opsonization and phagocytosis

- Opsonization ; coating particles for phagocytosis and the molecules coating the particle are opsonins
- Phagocytes mediate their phagocytosis in three ways
 - By binding directly to antigens and ingesting them
 - By using C3b receptor to bind complement opsonized particles
 - By using Fc receptors to bind antibody opsonized antigens
-
-
- There are different kinds of FC receptors

⑧ Fc receptor	Affinity for Ig	Cell distribution	Function
FcγRI (CD64)	High ($K_d \sim 10^{-9}$ M); binds IgG1 and IgG3; can bind monomeric IgG	Macrophages, neutrophils; also eosinophils	Phagocytosis; activation of phagocytes
FcγRIIA (CD32)	Low ($K_d \sim 0.6-2.5 \times 10^{-6}$ M)	Macrophages, neutrophils; eosinophils, platelets	Phagocytosis; cell activation (inefficient)
FcγRIIB (CD32)	Low ($K_d \sim 0.6-2.5 \times 10^{-6}$ M)	B lymphocytes	Feedback inhibition of B cells
FcγRIIIA (CD16)	Low ($K_d \sim 0.6-2.5 \times 10^{-6}$ M)	NK cells	Antibody-dependent cellular cytotoxicity (ADCC)
FcεRI	High ($K_d \sim 10^{-10}$ M); binds monomeric IgE	Mast cells, basophils, eosinophils	Activation (degranulation) of mast cells and basophils

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FcγRIIb	low affinity	neutrophils	phagocytosis
FcμR	medium affinity	neutrophils	phagocytosis

Antibody mediated cellular cytotoxicity (ADCC)

- III to kill target cells □ NK cells use FC
- RI to kill helminthes □ Eiosinophills use FC

Complements

- complement system is collection of circulating and cell membrane proteins (C1-C9)
- activated by microbes or microbes bound to antibodies
- activation of complement proteins involves sequential proteolytic cleavage of these proteins leading to generation of effector molecules
- are synthesized by hepatocytes but also some amounts by macrophages GIT epithelial cells
- three pathways of activation
 - **Alternative** :when aa break down product of C3 hydrolysis by C3 convertase enzyme (C3b)to the microbe .when C3b forms covalent molecule with the microbe no further degradation .**FIRST** to act
 - **Classical**: triggered by activation of C-1 complex (C1q, 2 C1r, & 2 C1s). When C1q binds to IgM or IgG complexed with antigens or when it directly binds to the microbe while single IgM can initiate this pathway, multiple IgGs are needed. **THIRD** to act
 - **Lectin** : Homologous to classical pathway, but with the opsonin, MBL and ficolins instead of C1q. Activated by binding of MBL to mannose residue of pathogen surface, which activates MBL-associated serine proteases (MASP-1 & -2), can split C4 into C4a and C4b and C2 into C2a & C2b. (**SECOND** to act)
- all of the activation leads to the cleavage of C3 to C3a and C3b
- functions include
 - Opsonization-enhancing phagocytosis of antigens
 - Chemotaxis-attracting macrophages and neutrophils
 - Cell lysis-rupturing membrane of foreign cells

- Agglutination-clustering and binding of pathogenstogether (sticking)
- Initiating inflammatory reaction (C5a, C3a, C4a)
- Membrane attack complex MAC C5b, C6,C7,C8,C9)
- Clearance of immune complex :bound C3b binds to the Cr1 receptor on erythrocytes then gets cleared out in the spleen

Antibodies on special effector sites

- IgA offers protection in mucosal tissues .the secreted Ab is transported through the epithelial cells by IgA fc specific receptor called poly Ig receptor (recognized IgM)
- Neonatal immunity:Maternal IgG is transported across the placenta through neonatal Fc receptor (FcRn). maternal IgA and IgG in breast milk are ingested by nursing infant

There are many ways pathogens avoid being detected by the humoral immune system these include

- Antigenic variety
- Inhibition of complement inactivation

The hemolytic disease of the new born as we have learnt is when the fetal blood leaks to the maternal circulation and this is recognized by the mothers cell as antigen so the antibodies of the mother will attack the erythrocytes of the fetus thus hemolytic disease of the new born will occur.

Most vaccines trigger humoral immune response.

*COMPLEMENTS, GRAFTS AND
IMPLANTATION IMMUNOLOGY*

Complement and Infection

- ✓ The complement system is a collection of circulating and cell membrane proteins that play important roles in host defense against microbes and in antibody mediated tissue injury. (C1-C9)
- ✓ The term complement refers to the ability of these proteins to assist, or complement, the antimicrobial activity of antibodies.
- ✓ The CS activated by

1. **Microbes**; in the absence of antibody, as part of the innate immune response to infection, &

2. **Antibodies** attached to microbes, as part of adaptive immunity.

- ✓ The activation of complement proteins involves *sequential proteolytic cleavage* of these proteins and leads to the generation of *effector molecules* that participate in eliminating microbes in different ways.
- ✓ Activated complement proteins become covalently attached to the cell surfaces where the activation occurs, ensuring that activation is limited to the correct sites.
- ✓ The proteins and glycoproteins that constitute the system are synthesised by *hepatocyte*. Also produced by *macrophages*, *blood monocytes* & *epithelial cells* of the genitourinal and gastrointestinal tracts.

There are three major pathways

1. *Alternate*
2. *Lectin*
3. *classical pathway*

- ✓ The first two are initiated by microbes in the absence of antibody.
- ✓ The most abundant complement protein in the plasma, called C3, plays a central role in all three pathways

1. Alternate Pathway

- Is triggered when a breakdown product of C3 hydrolysis (3Cb), is deposited on the surface of a microbe.
- C3b is generated from C3 by a C3 convertase enzyme complex in the fluid phase that is rapidly inactivated by factor H and I.
- when C3b forms stable covalent bond with microbial proteins or polysaccharides it is protected from further degradation(C3b is prevented from binding stably to normal host cell)
- The microbe-bound C3b becomes a substrate for the binding of another protein called factor B, which is broken down by a plasma protease to generate the Bb fragment. This fragment remains attached to the C3b, and the C3bBb complex enzymatically breaks down more C3, functioning as the “alternative pathway C3 convertase.”
- The *C3bBb3b complex* functions as a *C5 convertase*, to break down the complement protein C5 and initiate the late steps of complement activation.

2. Classical Pathway

- Triggered when IgM or certain subclasses of IgG (IgG1 and IgG3 in humans) bind to antigens. (Activation of C-1 complex (C1q, 2 C1r, & 2C1s).
- Single IgM can initiate this pathway, while multiple IgGs are needed.
- can also occurs when C1q binds directly to surface of pathogen, & this in turn leads to conformational changes in C1q molecule, which leads to activation of 2C1r molecules.

- C1r is a serine protease and cleaves C1s another serine protease.
- C1r2s2 component splits C4 and then C2, producing C4a, C4b, C2a & C2b.
- C4b and C2a bind to form C4b2a complex, which promotes cleavage of C3 into C3a and C3b; C3b later joins C4b2a (C3 convertase) to make C5 convertase (C4b2a3b complex).

3. Lectin Pathway

- Initiated in the absence of antibody by the attachment of plasma mannose-binding lectin (MBL) to microbes.
- MBL is structurally similar to a component of C1 of the classical pathway and serves to activate C4.
- The subsequent steps are essentially the same as in the classical pathway

NB

- ❖ The net result of these early steps of complement activation is that microbes acquire a coat of covalently attached C3b.
- ❖ These pathways differ in how they are initiated, but once they are triggered, their late steps are the same.
- ❖ The late steps of complement activation are initiated by the binding of C5 to the C5 convertase, and the subsequent proteolysis of C5, generating C5b.
- ❖ The remaining components, C6, C7, C8, and C9, bind sequentially.
- ❖ This poly-C9 is called the membrane attack complex, and its formation is the end result of complement activation.

Functions of complement

- ❖ *Opsonization*-enhancing phagocytosis of antigens
- ❖ *Chemotaxis*-attracting macrophages and neutrophils

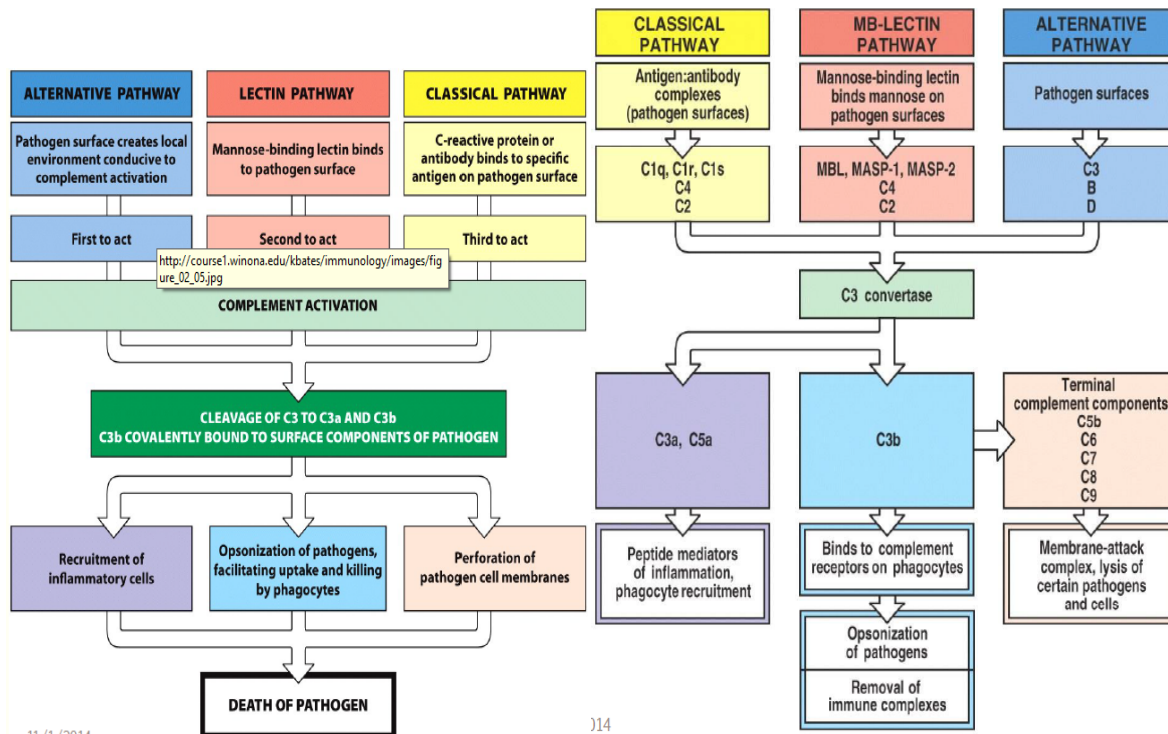
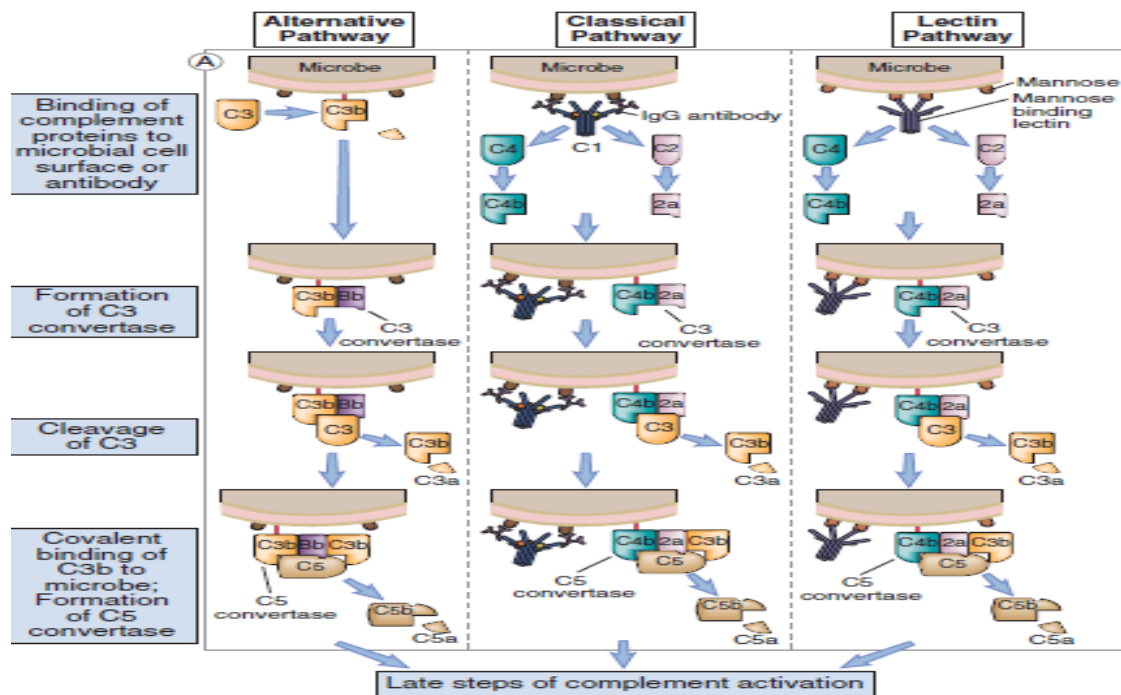
❖ **Cell lysis**-rupturing membrane of foreign cells □Agglutination-clustering and binding of pathogens together (sticking)

❖ **Agglutination**-clustering and binding of pathogens together (sticking)

Ⓒ Protein	Serum conc. (ug/mL)	Function
C1 (C1qr2s2)		Initiates the classical pathway; C1q binds to Fc portion of antibody; C1r and C1s are proteases that lead to C4 and C2 activation
C4	300-600	C4b covalently binds to surface of microbe or cell where antibody is bound and complement is activated C4b binds to C2 for cleavage by C1s C4a stimulates inflammation
C2	20	C2a is a serine protease functioning as an active enzyme of C3 and C5 convertases
Mannose binding lectin (MBL)	0.8-1	Initiates the lectin pathway; MBL binds to terminal mannose residues of microbial carbohydrates. A MBL-associated protease activates C4 and C2, as in the classical pathway.

Ⓑ Protein	Serum conc. (ug/mL)	Function
C3	1000-1200	C3b binds to the surface of a microbe where it functions as an opsonin and as a component of C3 and C5 convertases C3a stimulates inflammation
Factor B	200	Bb is a serine protease and the active enzyme of C3 and C5 convertases
Factor D	1-2	Plasma serine protease which cleaves Factor B when it is bound to C3b
Properdin	25	Stabilizes the C3 convertase (C3bBb) on microbial surfaces

Ⓑ Protein	Serum conc. (ug/mL)	Function
C5	80	C5b initiates assembly of the MAC C5a stimulates inflammation
C6	45	Component of the MAC: binds to C5b and accepts C7
C7	90	Component of the MAC: binds C5b, 6 and inserts into lipid membranes
C8	60	Component of the MAC: binds C5b, 6, 7 and initiates binding and polymerization of C9
C9	60	Component of the MAC: binds C5b, 6, 7, 8 and polymerizes to form membrane pores



Immunological bases of Graft -Rejection

- Graft rejection is mediated by the ***adaptive immune system***, because it shows specificity and memory and is dependent on lymphocytes.

Evidence	Conclusion
Prior exposure to donor MHC molecules leads to accelerated graft rejection	Graft rejection shows memory and specificity, two cardinal features of adaptive immunity
The ability to reject a graft rapidly can be transferred to a naive individual by lymphocytes from a sensitized individual	Graft rejection is mediated by lymphocytes
Depletion or inactivation of T lymphocytes by drugs or antibodies results in reduced graft rejection	Graft rejection can be mediated by T lymphocytes

FIGURE 10-6 Evidence indicating that the rejection of tissue transplants is an immune reaction. Clinical and experimental evidence indicates that rejection of grafts is a reaction of the adaptive immune system. MHC, major histocompatibility complex.

Graft rejection is classified into 3 based on ***clinical*** and ***pathological*** features

1. Hyper acute
2. Acute and
3. Chronic

1. Hyper acute rejection

- Occurs within minutes of transplantation and is characterized by thrombosis of graft vessels and ischemic necrosis of the graft.
- Hyper acute rejection is mediated by circulating antibodies, specific for antigens on graft endothelial cells, which are present before transplantation.
- These preformed antibodies may be natural IgM antibodies specific for blood group antigens, or they may be antibodies specific for allogeneic MHC molecules that are present because of exposure to allogeneic cells due to blood transfusions, pregnancy, or prior organ transplantation.

- Hyper-acute rejection is not a common problem in clinical transplantation, because every recipient is tested for blood type and for antibodies against the cells of the potential donor. (The test for antibodies is called a cross-match.)
- However, hyper-acute rejection is the major problem in *xenotransplantation* (transplantation of organ, tissue or cells between two different species)

2. Acute Rejection

- Occurs within days or weeks after transplantation and is the principal cause of early graft failure.
- Acute rejection is mediated mainly by T cells, which react against alloantigen in the graft.
- These T cells may be CTLs that directly destroy graft cells, or the T cells may react against cells in graft vessels, leading to vascular damage.
- Antibodies also contribute to acute rejection, especially the vascular component of this reaction, when injury to graft vessels is caused mainly by complement activation by the classical pathway.
- Current immunosuppressive therapy is designed mainly to prevent and reduce acute rejection by blocking the activation of allo-reactive T cells,

3. Chronic rejection

- An indolent form of graft damage that occurs over months or years, leading to progressive loss of graft function.
- Chronic rejection may be manifested as fibrosis of the graft and by gradual narrowing of graft blood vessels, called *graft arteriosclerosis*.

- In both lesions, the culprits are believed to be T cells that react against graft alloantigen and secrete cytokines, which stimulate the proliferation and activities of fibroblasts and vascular smooth muscle cells in the graft.

PREVENTION AND TREATMENT OF GRAFT REJECTION

- ✚ The mainstay of preventing and treating the rejection of organ transplants is immunosuppression, designed mainly to inhibit T cell activation and effector functions.

Graft-versus-host disease (GVHD)

- A disease occurring in bone marrow transplant recipients that is caused by the reaction of mature T cells in the marrow graft against alloantigen on host cells. The disease most often affects skin, liver, and intestines.
- The immune system reacts very strongly against allogeneic bone marrow cells, so that successful transplantation requires careful HLA matching of donor and recipient. If mature allogeneic T cells are transplanted with the marrow cells, these mature T cells can attack the recipient's tissues, resulting in this serious clinical reaction.

Transplantation immunology and immunosuppressive therapy

- Animals that are identical to one another (and grafts exchanged among these animals) are said to be *syngeneic*; animals (and grafts) of one species that differ from other animals of the same species are said to be *allogeneic*; and animals (and grafts) of different species are *xenogeneic*.

TRANSPLANTATION ANTIGENS

- The *antigens of allografts* that serve as the principal targets of rejection are proteins encoded in the *MHC* (major histocompatibility).
- Homologous MHC genes and molecules are present in all mammals; the human MHC is the human *leukocyte antigen (HLA) complex*.
- Every human being expresses six class I MHC alleles (one allele of HLA-A, -B, and -C from each parent) and at least six class II MHC alleles (one allele of HLA-DQ and -DP and one or two of -DR from each parent, and some combinations of these)
- MHC genes are highly polymorphic; it is estimated that in the population there are at least 350 alleles of HLA-A genes, 620 alleles of HLA-B genes, 400 alleles of DR genes, and 90 alleles of DQ genes. Because these alleles can be inherited and expressed in many different combinations, *every individual is likely to express some MHC proteins that appear foreign to another individual's immune system, except in the case of identical twins*.
- All of the MHC molecules may be targets of rejection, although HLA-C and HLA-DP have limited polymorphism and probably are of minor significance.
- The reason why individuals react against MHC molecules of other individuals is now understood quite well. T cell antigen receptors (TCRs) have evolved to recognize MHC molecules, which is essential for surveillance of cells harboring infectious microbes. In each individual, all CD4⁺ T cells and CD8⁺ T cells are

selected during their maturation to recognize peptides displayed by that individual's (self) MHC molecules. This selection is the basis for the “*self MHC restriction*” of T lymphocytes, a fundamental property of T cells.

- As a result of positive selection in the thymus, mature T cells that have some affinity for self MHC molecules survive, and many of these will have high affinity for self MHC displaying foreign peptides. Allogeneic MHC molecules, containing peptides derived from the allogeneic cells, may look like self MHC molecules plus bound foreign peptides. Therefore, recognition of allogeneic MHC molecules in allografts is an example of an immunologic cross-reaction.
- The process of negative selection in the thymus eliminates cells that strongly recognize self MHC, but there is no opportunity to selectively eliminate T cells whose TCRs have a high affinity for allogeneic MHC molecules, which are never present in the thymus.
 - Although MHC proteins are the major antigens that stimulate graft rejection, other polymorphic proteins also may play a role in rejection. Non-MHC antigens that induce graft rejection are called *minor histocompatibility antigens*, and most of them are allelic forms of normal cellular proteins that happen to differ between donor and recipient. The rejection reactions that minor histocompatibility antigens elicit usually are not as strong as reactions against foreign MHC proteins.

INDUCTION OF IMMUNE RESPONSES AGAINST TRANSPLANTS

- The induction of T cell-mediated immune responses against tissue transplants has to overcome the same problem as in responses against tumors: Because a graft may contain many cell types, often including epithelial and connective tissue cells,

- T cells in the recipient may recognize donor alloantigen in the graft in different ways, depending on what cells in the graft are displaying these alloantigen.

1. **Direct recognition**: - of allosteric occurs when T cells bind directly to intact allogeneic major histocompatibility complex (MHC) molecules on professional antigen-presenting cells (APCs) in a graft.

- Direct recognition stimulates the development of alloreactive T cells (e.g., CTLs) that recognize and attack cells of the graft.

2. **Indirect alloantigen recognition**: - occurs when allogeneic MHC molecules from graft cells are taken up and processed by recipient APCs, and peptide fragments of the allogeneic MHC molecules are presented by recipient (self) MHC molecules.

- Much like that for recognition of any foreign antigen.
- Donor alloantigen are processed and presented by the self MHC molecules on recipient APCs.

The mixed lymphocyte reaction (MLR)

- It is an in vitro model of T cell recognition of alloantigens.
- In this model, T cells from one individual are cultured with leukocytes of another individual, and the responses of the T cells are assayed.
- The magnitude of this response is proportional to the extent of the MHC differences between these individuals and is a rough predictor of the outcomes of grafts exchanged between these individuals.

Although much of the emphasis on allograft rejection has been on the role of T cells, it also is clear that *alloantibodies* contribute to rejection. Most of these antibodies are helper T cell-dependent high-affinity antibodies.

Immunosuppressive therapy

- *Immunosuppressive* or *anti-rejection* medications are drugs that inhibit or prevent activity of the immune system.
- They are used in immunosuppressive therapy to prevent
 - ✓ rejection of transplanted organs and tissue (bone marrow, heart , kidney)
 - ✓ treat autoimmune diseases (rheumatoid arthritis, multiple sclerosis)
 - ✓ treat some other non-auto immune inflammatory disease (asthma)
- ❖ A common side effect of many immunosuppressive drugs is *immunodeficiency*, because the majority of them act non-selectively, resulting in increased susceptibility to infections and depressed immuno-surveillance. Other side effects includes hypertension, dyslipidemia, hyperglycemia etc...
- ❖ Immunosuppressive drugs can be classified into 5 groups
 -  glucocorticoids
 -  cytostatic
 -  antibodies
 -  immunophilins acting drugs
 -  other drugs

*HYPERSENSITIVITY, IMMUNODEFICIENCY,
AUTOIMMUNE DISEASE AND AMYLOIDOSIS*

Hypersensitivity reaction

- Pathogenesis – in certain individuals repeated exposure to specific antigen leads to development of adverse reaction by:
 - Release of vasoactive and spasmogenic inflammatory mediators – type 1
 - Binding of antibody to antigen on cell surface – type 2
 - Formation of antigen antibody complex capable of activation of complement system – type 3
 - Sensitization of t-lymphocytes after reaction with antigen leading to granulomatous inflammation – type 4

Type 1 hypersensitivity reaction

- Develop rapidly (within minutes) following exposure to the antigen in a previously sensitized individual,
- Based on a mast cell participation & Mediated by IgE in most instances
- Type I hypersensitivity reaction may cause a localized or a systemic reaction.

Localized reaction: If the allergen is confined to the site of contact, localized edema and inflammation typically develop (e.g., hives and rhinitis). This reaction causes only minor discomfort, but it may also be life threatening if it involves the larynx or the bronchial tree (e.g., bronchial asthma).

Systemic reaction: A life-threatening anaphylactic shock may develop, especially if the antigen was introduced intravenously (e.g., antibiotic).

- These reactions involve the interaction of three cell types:
B cells: Virgin B cells stimulated by antigen (allergen) transform into IgE-secreting plasma cells.
TH2 lymphocytes: Helper T lymphocytes produce IL-4, which is necessary for stimulation of IgE-secreting B cells.
Mast cells: These cells bind IgE to their surface and act as effector cells.
- First exposure leads to a sensitization and priming of the T and B cells, which interact and produce IgE that binds to the surface of mast cells in tissues.
- Upon second exposure to the same antigen, cross-linking of IgE on the mast cells occurs.
- Cell surface changes stimulate two sets of reactions
 - **Immediate reaction:** This involves a rapid release of mediators stored in mast cell granules.
 - **Delayed reaction:** This reaction is mediated by substances that need to be synthesized, and their release occurs several hours after the initiation of the immune reaction.

Antigen-triggered degranulation of mast cells leads to a release of preformed (primary) mediators of inflammation:

- **Biogenic amines:** Histamine, the primary mediator of this group, will increase vascular permeability (edema) and stimulate smooth muscle contraction (bronchospasm) and secretion of mucus or gastric juices.
- **Enzymes:** Hydrolytic enzymes (e.g., chymase and tryptase) contribute to inflammation by activating kinins and the complement system. Fragments of complement proteins have a chemotactic action and recruit other cells to the site of inflammation.

- **Proteoglycans:** The most important is heparin, which has an anticoagulant action.

Secondary mediators released 6 to 12 hours later, include:

- **Lipid-derived mediators:** Antigen binding to mast cells triggers an activation of phospholipase A₂ that generates arachidonic acid from intracellular lipids. Arachidonic acid is further metabolized into leukotrienes and prostaglandins. Platelet-activating factor (PAF) is also formed.
- **Cytokines:** This response is characterized by the recruitment of inflammatory cells (neutrophils and eosinophils) that sustain the inflammation, causing further damage in the affected tissue.

The symptoms depend on the route of exposure, dose of antigen, and the target organ sensitivity

- **Systemic allergen presentation:** Typically this reaction occurs upon injection of drugs, hormones, or antisera into a previously sensitized person. Clinically, an anaphylactic shock develops, which is characterized by hypotension due to widespread vascular dilatation, widespread edema, difficulty breathing due to laryngospasm, laryngeal and pulmonary edema, and cardiac dysrhythmia.
- **Localized reactions:** Antigen entering the body through the skin, by inhalation, or through ingestion evokes a reaction usually limited to one organ system.
 - **Upper respiratory tract reactions:** Seasonal upper respiratory allergies caused by pollens (>10 mm) evoke hay fever. Similar

but prolonged allergic rhinitis may result from sensitization to dust mites. In both instances, there is nasal congestion, watery discharge, itching, sneezing, and cough.

- **Gastrointestinal reactions:** Ingestion of allergens may lead to diarrhea, malabsorption syndrome, and protein-losing enteropathy. Massive hypoproteinemia may even cause shock.
- **Bronchopulmonary reactions:** Asthma is an example of hypersensitivity of small airways to inhaled allergens. It presents with bronchospasm causing wheezing, hypersecretion of mucus, and coughing. Death due to asphyxia may result from extreme bronchoconstriction and plugging of bronchi with mucus (status asthmaticus).
- **Skin reactions:** Atopic dermatitis, a common disease of childhood, is caused by prolonged exposure to and sensitization by foreign antigens.

Type 2 hypersensitivity reaction

- Antigen is present on the surface of the target cell. It may be intrinsic antigen (e.g., Rh D blood antigen) or antigen adsorbed on the cell surface from the environment (e.g., drugs acting as haptens).
- The antibody interaction with the antigen elicits three reactions (Fig. 4-3):
 - **Complement-activated cell lysis:** Binding of antibody (primarily IgM) to the antigen on the cell surface leads to an activation of complement-activated cell lysis and the formation of the cytolytic

membrane attack complex (MAC), which destroys cells (e.g., acute hemolytic transfusion reaction).

- **Antibody-dependent cell-mediated cytotoxicity (ADCC):** Binding of low-density (primarily IgG) antibodies that act as opsonins is followed by phagocytosis of damaged cells. ADCC typically mediates autoimmune hemolytic anemia and thrombocytopenia.
- **Antibody-mediated cell dysfunction:** Binding of stimulatory antibodies to receptors for thyroid-stimulating hormone (TSH) on thyroid cells may stimulate hypersecretion of thyroid hormones in Graves' disease. Binding of anti-acetylcholine receptor antibodies at the neuromuscular junction may block nerve transmission and the cause of muscle weakness in myasthenia gravis.

Important diseases caused by type II hypersensitivity cytotoxic reactions.

- **Transfusion reaction:** This is caused by a mismatched blood transfusion. An infusion of blood of an A blood group donor into a B blood group recipient will result in a massive hemolytic reaction mediated by the natural anti-A blood group antigen in the recipient.
- **Hemolytic disease of the newborn (erythroblastosis fetalis):** This reaction occurs transplacentally and is mediated by maternal IgG antibodies to the fetal Rh_p positive red blood cells (RBCs). It occurs only in Rh mothers who have been immunized to Rh blood group antigen during the first pregnancy by an Rh_p fetus.

- **Autoimmune hemolytic anemia:** This disease results from the production of anti-RBC antibodies that bind to RBC and cause hemolysis.
- The causes of such an immune reaction are not known.
- **Pemphigus vulgaris:** This blistering skin disease is caused by antibodies to the proteins forming the intercellular junctions (desmosomes) of the epidermis.
- **Goodpasture syndrome:** This pulmonary-renal disease is caused by antibodies that react with type IV collagen in the glomerular and pulmonary vascular basement membranes.
- One could call these antibodies cytotoxic, but the term membranotoxic seems more appropriate.

Type 3 hypersensitivity reaction

- Binding of antibody to the antigen forms complexes that are not cleared by the reticuloendothelial system but instead are deposited in tissues where they elicit an inflammatory reaction through activation of complement & Complexes may be formed locally in the tissue or in the circulation and then deposited in various tissues.
- Either way, aggregated Fc regions of IgG antibodies activate the complement pathway, leading to increased vascular permeability, activation of neutrophils, and tissue necrosis.
- Antigens capable of inducing type III hypersensitivity reaction may be exogenous (e.g. Streptococcal antigens, hepatitis B virus, and heroin) or endogenous (e.g., immunoglobulins and nuclear antigens).

Most important diseases caused by type III hypersensitivity reaction.

- **Systemic lupus erythematosus (SLE):** This disease is caused by circulating immune complexes that deposit in many organs, most often the skin, joints, and kidneys. Circulating blood cells are also often affected, resulting in anemia, leukopenia, and thrombocytopenia.
- **Polyarteritis nodosa:** This disease typically affects medium-sized arteries of internal organs, skin, eye, or peripheral nerves in an unpredictable manner.
- **Acute post-streptococcal glomerulonephritis:** This kidney disease typically follows a streptococcal throat infection and the formation of circulating anti-streptococcal antibodies, which deposit in the glomerular basement membranes.
- **Membranous nephropathy:** In some forms of this immune-mediated glomerulopathy, there are circulating antigen-antibody complexes that deposit on the epithelial side of the glomerular basement membrane. The disease typically presents as nephrotic syndrome and massive proteinuria.
- **Serum sickness:** Injection of foreign serums may cause the formation of circulating immune complexes that are deposited in many organs. Most often it presents with a skin rash.

Type 4 cell mediated hypersensitivity reaction

- Type IV or delayed type hypersensitivity (DTH) response.
- In this response, antigen-presenting cells in the skin (dendritic Langerhans cells) internalize self-proteins that have exogenous molecules bound to them (haptenated self-proteins), which renders these self-proteins immunogenic.
- These complexes are displayed on class II MHC molecules, which bind to TH cell receptors (TCRs).
- This results in activation of sensitized TH1 (TDTH) cells, release of cytokines, and subsequent activation of tissue macrophages (via IFN- γ) and CD8 β T cells (via IL-2).
- The release of lytic enzymes by the macrophages, as well as epidermal damage by CTLs, results in the erythema, weepiness, and vesicles that characterized this patient's reaction to benzocaine.
 - Primarily mediated through the interaction of TH1 helper CD4 β and cytotoxic CD8 β T lymphocytes.
 - A classic example is tuberculin reaction: In a previously sensitized individual (exposed to *Mycobacterium tuberculosis*), intracutaneous injection of purified protein derivate (PPD) of *M. tuberculosis* cultures will lead to delayed (8-12 hours) redness and induration at the site of injection. It reaches the peak in 24 to 72 hours, after which it slowly subsides.

- Characterized by the accumulation of mononuclear cells (predominantly CD4⁺ T cells) around small vessels (cuffing).
- If the antigen is non-degradable, perivascular lymphocytes are replaced by macrophages that resemble epithelium (epithelioid macrophages) and form a granuloma.
- Most important diseases caused by type IV hypersensitivity reaction. Contact dermatitis (poison ivy), Tuberculosis, Sarcoidosis, Guillain Barre syndrome and multiple sclerosis...

Amyloidosis

- Amyloid is a proteinaceous substance deposited between cells in various tissues and organs in a variety of clinical settings.
- The name derives from its starch-like staining properties (PAS [periodic acid shift] positive).
- Amorphous, eosinophilic, hyaline extracellular substance is seen under the microscope.
- Amyloids are chemically diverse and have typical physical properties.
- All amyloids, irrespective of their biochemical composition, form non-branching long fibrils (7.5–10 nm).
- By electron diffraction, they are arranged into β -pleated sheaths.
- Congo red staining is typical. Under polarized light, Congo red-stained amyloid fibrils show an apple green birefringence.

Three most common biochemical forms of amyloid?

- **AL:** This form of amyloid is formed from immunoglobulin light chains. It is found in multiple myeloma.
- **AA:** This amyloid-associated protein is synthesized by the liver (in the form of the precursor serum amyloid A [SAA]). The form is seen in tissues of patients harboring chronic suppurative infections (e.g., bronchiectasis and osteomyelitis).
- **Ab:** This amyloid is found in Alzheimer disease (b-amyloid protein).
- Amyloidosis is a group of diseases characterized by a deposition of amyloid in various organs.
- The deposition of amyloid may occur as a primary disease (e.g., in multiple myeloma) or in a secondary form during the course of chronic inflammatory or other diseases.
- The disease may be limited (localized) to a single organ, or it may be systemic. Some forms may be hereditary.
- **Primary amyloidosis:** Systemic deposition of AL-type amyloid, Associated with plasma cell neoplasia and B-cell lymphoma, Monoclonal gammopathy (whole immunoglobulins or only light chains) found by electrophoresis of serum
- **Secondary systemic amyloidosis:** Underlying disease (e.g., tuberculosis, chronic osteomyelitis, rheumatoid arthritis, and cancer) or condition (heroin abuse). Widespread deposition of AA protein in many organs

Amyloidosis of aging

- Systemic deposition of amyloid in elderly patients (seventh and eighth decades). Frequent involvement of the heart (cardiac amyloidosis)

Most severely affected organs in systemic amyloidosis

- **Kidneys:** Both kidneys are enlarged, pale, and waxy. Glomerular mesangial, interstitial, and vascular wall depositions of amyloid are found.
- **Spleen:** Splenomegaly presents with either nodular (follicular) depositions (sago spleen) or map-like (red pulp) depositions (lardaceous spleen).
- **Liver:** Hepatomegaly, extracellular amyloid with pressure atrophy of hepatocytes.
- **Small blood vessels:** Many organs show vascular deposits of amyloid, and, accordingly, the diagnosis may be made by biopsy of tissue from the fat pad, gingiva, or rectum.

Auto immune diseases

- Autoimmune diseases are caused by an immune reaction against self-antigens. They may involve a single organ or be multisystemic.

How does autoimmunity develop?

- **Failure of peripheral tolerance:** Activation of anergic clones later in life occurs because tissue damage has caused this antigen to present with costimulatory B-7 on APCs.

- **Release of sequestered antigens:** Antigens normally unavailable to the cells of the immune system become available and unmasking of cryptic antigens in the context of inflammation (to upregulate costimulatory pathways) occurs.
- Genetic predisposition may take several forms. Expression of certain HLA alleles confers higher susceptibility to the development of autoimmunity.
- Mutations (e.g., in Fas–Fas ligand) or polymorphisms in the genes responsible for the tolerance may lead to susceptibility to autoimmunity.
- Infections by viruses and other microbes may cause autoimmunity because of cross-reactivity, tissue necrosis (resulting in activation of anergic clones), or general stimulation (caused by the so-called super antigens) of the immune system.

Why do some autoimmune diseases affect single organs, whereas others are systemic?

- The outcome of an autoimmune reaction depends on the nature of the antigen.
- If the antigen is restricted in its expression to a single organ (e.g., TSH receptor on the thyroid gland), the autoimmune reaction will be limited to the cells of that organ.
- When more organs express the same antigen, more widespread disease ensues (multi-organ disease).
- Furthermore, several types of immune reactions against different self-antigens may be seen in systemic autoimmune disease.

SLE/systemic lupus erythematosus/

- SLE is an example of a systemic autoimmune disease capable of affecting virtually every organ in the body.
- It is characterized by a wide array of antibodies against self-antigens, neither organ- nor species-specific.
- SLE affects young women (in their 20s or 30s).
- A person is considered to suffer from SLE if at least 4 of 11 clinical and laboratory criteria are identified simultaneously or cumulatively at any interval of observation.
- These criteria are established by the American College of Rheumatology:
 1. Malar (“butterfly”) rash over the face
 2. Discoid rash
 3. Photosensitivity (rash as a result of sun exposure)
 4. Oral ulcers
 5. Arthritis (nondeforming polyarthritis)
 6. Serositis (pleuritis and/or pericarditis)
 7. Renal disorder (proteinuria of >0.5 g/day; cellular casts as signs of glomerulonephritis)
 8. Neurologic disorder (seizures and/or psychosis)
 9. Hematologic disorder (leukopenia, lymphopenia, thrombocytopenia, and hemolytic anemia)
 10. Immunologic disorder (antibodies to native DNA or Smith antigen [anti-Sm]; antiphospholipid antibodies)
 11. Antinuclear antibody (in the absence of drug-induced lupus)

- During the flare-up of SLE, patients usually have fever, malaise, weight loss, and systemic signs of inflammation.
- Raynaud phenomenon, presenting as blanching of fingers and toes, is a common finding.
- Some patients have nonbacterial, sterile endocarditis (Libman–Sacks endocarditis), and occasionally some have neurologic and eye manifestations.
- Renal failure and infections are the most common cause of death.

Are the antibodies against nuclear antigens (ANAs) diagnostic of SLE?

- Immunofluorescence test for ANAs is positive in almost all patients with SLE, but it may also be positive in many other autoimmune diseases, such as systemic sclerosis, Sjögren syndrome, and dermatomyositis.
- ANAs are also found in 5% to 15% of otherwise healthy people, and their incidence increases with age.
- This test has high sensitivity but low specificity for SLE.

What is the best serologic test for diagnosing SLE?

- Antibodies to double-stranded DNA and to anti-Sm are virtually diagnostic of SLE but are seen in only 20% to 30% of patients.
- These tests have high specificity but low sensitivity for SLE.

Why do patients with SLE frequently have positive VDRL and RPR tests for syphilis?

- Many SLE patients (40%–50%) have antiphospholipid antibodies, which often react with the cardiolipin antigen used as reagent in these serologic tests for syphilis.
- This cross-reactivity may account for the false-positive syphilis tests in SLE patients.

What is lupus anticoagulant?

- SLE patients may have coagulation disorders that are often features of an antiphospholipid antibody syndrome.
- Antibodies found in this syndrome may lead to a prolongation of activated partial thromboplastin time (aPTT).
- Lupus anticoagulant can be demonstrated in the blood of these patients.
- Paradoxically, patients with antiphospholipid antibody syndrome suffer from recurrent venous and arterial thromboses.

Does maternal SLE pose any danger to the offspring?

- SLE can be passed on to the offspring.
- Familial, twin, and ethnic studies suggest a genetic predisposition to SLE, which is inherited as a multifactorial disease.
- The incidence of the disease among infants born to mothers with SLE is in the range of 2% to 7%.

- Neonatal lupus syndrome is a rare condition caused by the passive transplacental transfer of IgG autoantibodies from the mother with autoimmune disease.
- Skin rash, hematologic cytopenias, hepatosplenomegaly, myocarditis, and serositis are typically found.
- Symptoms mostly disappear within 6 to 8 months.

What are the most characteristic microscopic findings in SLE?

- SLE is characterized by widespread deposition of immune complexes in tissues.
 - Skin: Granular deposits of immune complexes along the epidermal-dermal basement membrane.
 - Kidneys: Deposits of immune complexes are found in the glomeruli, in the blood vessels, and in the basement membranes of the tubules.
 - Heart: Immune complex deposits are seen on the valves, most often on the mitral valve.

What is the range of renal involvement in SLE?

- Nearly all patients with SLE have some signs of renal disease. The severity of the renal injury varies, and the kidney biopsy may be categorized into five classes:
 - Class I: Normal biopsy (rare).
 - Class II: **Mesangial lupus glomerulonephritis**. It is characterized by the expansion of the intercapillary matrix and a mild increase in the number of mesangial cells.

- **Class III: Focal proliferative glomerulonephritis.** It is characterized by segmental proliferation of endothelial and mesangial cells associated with neutrophilic infiltration in less than 50% of all glomeruli.
- **Class IV: Diffuse proliferative glomerulonephritis.** It is the most severe form of renal SLE.
 - The majority of glomeruli show global hyper-cellularity, sometimes with crescents filling the urinary Bowman space.
- **Class V: Membranous glomerulonephritis.** It is principally characterized by the thickening of the capillary walls and a granular sub-epithelial deposition of immune complexes along the glomerular basement membranes.

Cardiac pathology of SLE.

- Pericarditis (most common)
- Endocarditis (Libman-Sacks)
- Nonspecific myocarditis
- Accelerated coronary atherosclerosis

Sjögren syndrome

- Sjögren syndrome is caused by an autoimmune destruction of lacrimal glands, causing dry eyes (xerophthalmia), and salivary glands, causing dry mouth (xerostomia).
- It may occur as an isolated disorder (primary Sjögren syndrome) or as secondary Sjögren syndrome associated with other autoimmune diseases (RA, SLE, and scleroderma).
- The following are typical features:
 - It affects mostly 35- to 45-year-old women (90% of patients are women).

- Antibodies against ribonucleoproteins (SS-A/Ro and SS-B/La) are found in approximately 80% of patients.
- Salivary and lacrimal glands are infiltrated with lymphocytes and enlarged.
- Patients with Sjögren syndrome are at high risk for developing lymphoma.

Scleroderma

- Scleroderma, also known as systemic sclerosis, is a systemic disease characterized by microvascular injury and excessive fibrosis associated with signs of cellular and humoral autoimmunity affecting skin and many internal organs.
 - An unknown triggering event leads to activation of lymphocytes.
 - Cytokines produced by activated lymphocytes stimulate fibroblasts to produce collagen, resulting in fibrosis.
 - An injury of the microvasculature leads to ischemic tissue injury.
 - Autoantibodies are common and point to an immunologic nature of the disease.
- ANAs (50%) are of low specificity and found in other autoimmune diseases.
- Antibodies to DNA topoisomerase I (anti-Scl 70) found in 50% of patients are highly specific for systemic sclerosis.
- Antibodies to centromere (30%) are less specific but are found in 90% of patients who have the CREST subtype of systemic sclerosis (see question 46).
- *Pathologic findings in scleroderma*
 - **Skin:** Atrophy and perivascular lymphocytic inflammation with edema followed by fibrosis, loss of skin appendages, and hyaline thickening of arterioles and capillaries.

- **Gastrointestinal system:** Esophagus and, to a lesser degree, small intestines are involved and show muscle atrophy and progressive fibrosis.
- **Musculoskeletal system:** Synovitis and myositis.
- **Kidneys:** The most striking changes involve interlobar arteries and arterioles. Intimal thickening with narrowing of the lumina with myxoid degeneration and fibrinoid necrosis.
- **Lungs:** Fibrosis and various degrees of vascular wall thickening.

Inflammatory myopathies.

- This term comprises several autoimmune diseases characterized by inflammation of skeletal muscles (myositis) and muscle weakness.
- Autoantibodies, like ANAs, are found in 50% of patients but are of low specificity.
- Jo-1 antibody (antibody to histidyl-t-RNA synthetase) is more specific but is found only in 25% of patients.
- The most important entities are:
 - Dermatomyositis
 - Polymyositis
 - Inclusion body myositis

Difference between dermatomyositis and polymyositis?

- **Polymyositis** is an autoimmune disease involving the skeletal muscles.
- It may occur in an isolated form or in conjunction with skin lesions.
- In this case, the syndrome is called **dermatomyositis**.
- Common to both diseases are the following features:
 - Preferential involvement of proximal parts of the extremities
 - Pain and symmetrical proximal muscle weakness
 - Histopathologic findings include infiltrates of lymphocytes and macrophages and muscle fiber injury
 - Creatinine kinase (CK) is elevated in serum.
- Patients affected by dermatomyositis also show typical skin lesions.
- Dermatomyositis of childhood is associated with heliotrope rash (purple reddish discoloration, most prominently seen on the upper eyelids).

Immunodeficiency

Define immunodeficiency.

- Quantitative or qualitative defects of the immune system.
- May involve the cells and components of:
- Natural defense mechanisms (e.g., defects of the phagocytic and complement-mediated defense)
 - Acquired immunity involving the T and B lymphocytes &
 - Associated with an increased susceptibility to infection and lymphoproliferative diseases.

Classify immunodeficiencies.

- **Primary:** This includes many congenital, mostly genetic deficiencies of one or more components of the immune system.
- **Secondary:** This includes numerous acquired deficiencies of one or more components of the immune system caused by infection, malnutrition, drugs, irradiation, cancer, or autoimmunity.

Characteristic manifestations of primary immunodeficiency's?

- Early onset, usually between 6 months and 2 years of age
- Recurrent infections

How are primary immunodeficiencies classified?

- B-cell deficiencies:
 - X-linked agammaglobulinemia (Bruton type)

- Common variable immunodeficiency
- Isolated IgA deficiency
- T-cell deficiencies:
 - Hyper-IgM syndrome
 - DiGeorge syndrome
- Severe combined immunodeficiency

Characteristics of X-linked agammaglobulinemia of Bruton?

- Failure to assemble complete immunoglobulin molecules due to a mutation in the btk (Bruton tyrosine kinase) gene located on the Xq21.2-22 chromosome
- Absent or markedly decreased concentration of all classes of immunoglobulins (agammaglobulinemia)
- Affects boys (X-linked disease)
- Symptoms appear after 6 months of age.
- Typically, there is an increased incidence of otitis media, and skin and respiratory infections caused by *Haemophilus influenzae*, *Streptococcus pneumoniae*, or *Staphylococcus aureus*.
- During the first 6 months, the infant is protected by maternally acquired antibodies and the infections appear only after these antibodies have been cleared from the infant's body.

Characteristics of common variable immunodeficiency?

- Late-childhood onset of hypogammaglobulinemia and recurrent infections
- Boys and girls affected equally

- High frequency of malignancies (gastric cancer and lymphoma) later in life

Characteristics of isolated IgA immunodeficiency?

- Isolated IgA deficiency is the most common congenital immunodeficiency, found in 1:850 people.
- Many of these men and women are asymptomatic.
- Because IgA is the major immunoglobulin found in mucosal secretions (respiratory, gastrointestinal, and urogenital tract), IgA deficiency is characterized by recurrent sinusitis, pulmonary and urinary infections as well as recurrent diarrhea.

Pathogenesis of X-linked hyper-IgM syndrome?

- Affected people have high titers of IgM and IgD antibodies but low titers of IgG, IgA, and IgE antibodies, indicating that there is a defect in isotype switching.
- The primary defect is in CD4⁺ T cells, which are responsible for production of cofactors necessary for stimulation of B-cell and isotype switching.
- The cause is a mutation of X-chromosome gene encoding CD40L. This protein expressed on CD4⁺ lymphocytes acts as a ligand binding to a B-cell surface receptor (CD40).
- Binding of this ligand is important for the activation of B cells.
- Patients suffer from pyogenic infections and are susceptible to infection with *Pneumocystis jiroveci*.

DiGeorge syndrome?

- T-cell deficiency due to the abnormal development of thymus
- Part of CATCH 22 syndrome: cardiac abnormality, T-cell deficit, cleft palate, hypocalcemia;
- syndrome is caused by a chromosome 22q11 deletion

Characteristics of severe combined immunodeficiency (SCID)?

- Prominent susceptibility to severe, recurrent fungal (*Candida albicans*), viral, and bacterial infections.
- Both humoral (antibodies) and cellular (T cells) immune responses severely reduced.
- X-chromosome linked (boys are typically affected); mutation involves the gene encoding cytokine receptors

HEMATOLOGICAL MALIGNANCIES

Hematologic malignancies

- **Leukemia** are tumors that primarily involve the bone marrow with spillage of neoplastic cells into the peripheral blood **lymphomas**, tumors that produce masses in involved lymph nodes or other tissues.
- **Plasma cell tumors**, the *plasma cell dyscrasias*, usually present within the bones as discrete masses and cause systemic symptoms related to the production of a complete or partial monoclonal immunoglobulin.

Leukemia

- Leukemia is a general term for a group of malignancies of either lymphoid or hematopoietic cell origin.
- The number of circulating leukocytes is elevated.
- The bone marrow is diffusely infiltrated with leukemic cells, therefore there will be decrease in normal leukocyte red cell, and platelet production can result in anemia, infection, or hemorrhage.
- It can be acute or chronic

1. Acute leukemia

- Blasts (immature and large cells) are characteristically predominant in bone marrow and peripheral blood.
- Bone marrow suppression occurs and accounts for the major clinical manifestation of acute leukemia.
- Suppression of normal hematopoietic stem cells, results in anemia, leukopenia, thrombocytopenia
- Acute leukemia occur most often in children.

- If Philadelphia chromosome occurs in acute leukemia is associated with poorer prognosis
- Clinical features; **abrupt onset** and ssx are related to cytopenias
- SSX include Fatigue, Infections, fever, Bleeding, Bone pain and tenderness, Generalized lymphadenopathy, splenomegaly, and hepatomegaly,
- CNS manifestations (meningeal spread)
- Lab findings include decrease in amount of hemoglobin, increase in total leukocyte count, decrease in platelet count and sometimes there is pancytopenia with few blast cells in the blood “**aleukemic leukemia**”
- Acute leukemia is subdivided in to ALL and AML based on the phenotype of the blasts

a) Acute lymphoblastic leukemia (ALL)

- Most common malignancy in children (80% of childhood malignancies)
- **Lymphoblasts** in z blood and bone marrow is characteristic
- Most responsive form of acute leukemia
- Sub-classified in to B-ALL and T-ALL

I. B-ALL

- In children (< 6 yrs) and young adults, M > F
- Most excellent response to therapy
- Prognosis depends on cytogenic abnormalities, could be good (in children) or poor (in adults)

II. T - ALL

- Usually presents in **T**eenagers as a mediastinal (**T**hymic) mass
- M > F
- Thymus, liver, spleen and lymph node are involved

Prognosis of ALL

- Most can be cured (especially children 2-10 years)
- Factors correlated with worse outcomes include male sex, age younger than 2 or older than 10, and a high leukocyte count at diagnosis, t(9,22)

b) Acute myeloid leukemia (AML)

- Is a neoplastic accumulation of myeloblasts
- Crystal aggregates of myelo-peroxidase (MPO) may be seen as **Auer rods**
- Affects older adults (mean age 50 years)
- Auer rods have the potential to activate coagulation cascade therefore there is increased risk of DIC
- Splenomegaly and lymphadenopathy are less prominent than in ALL
- Includes
 - **Acute promyelocytic leukemia** - t(15;17), RAR is disrupted and respond to vitamin A treatment
 - **Acute monocytic leukemia** - proliferation of monoblasts. Characteristically there is inflammation of the gums
 - **Acute megakaryoblastic leukemia** - proliferation of megakaryoblasts
- AML may also arise from pre-existing dysplasia for example prior exposure to chemotherapy or radiotherapy, tend to develop myelodysplastic syndrome.

Myelodysplastic syndrome (MDS)

- Cytopenias with hyper cellular bone marrow
- Age 50 - 70 yrs
- Defective bone marrow function results in infection, anemia and hemorrhage
- In some patients MDS may progress from aplastic anemia
- Poor response to chemotherapy

2. CHRONIC LEUKEMIA

- Neoplastic proliferation of **mature** circulating lymphocytes; characterized by a high WBC count

- Usually insidious in onset and seen in older adults

A. Chronic lymphocytic leukemia (CLL)/ small lymphocytic lymphoma (SLL)

- If the peripheral blood lymphocytosis exceeds 4000 cells/mm^3 , the patient is diagnosed with CLL; if not, a diagnosis of SLL is made otherwise are similar
- Neoplastic B cells, suppress normal B-cell function resulting in hypogammaglobinemia
- The predominant cells are compact, small, resting lymphocytes with dark-staining round nuclei, scanty cytoplasm, and little variation in size.
- Complications include

Hypogammaglobinemia—most common cause of death in CLL.

Autoimmune hemolytic anemia

Richter transformation— marked clinically by an enlarging lymph node or spleen

B. Hairy cell leukemia

- Neoplastic proliferation of **mature B cells** characterized by **hairy cytoplasmic processes**
- **CD11c and CD103**; these two antigens are not present on most other B-cell tumors, making them diagnostically useful.
- Occur in older males
- **Clinical features** include
 - Lymphadenopathy is usually absent.

- splenomegaly (due to accumulation of hairy cells in red pulp)
 - "dry tap" on bone marrow aspiration (due to marrow fibrosis)
 - Pancytopenia.
- ❖ extremely sensitive to chemotherapeutic agents, particularly purine nucleosides

C. Adult T cell lymphoma/leukemia (ATLL)

- Neoplastic proliferation of mature CD4⁺ T cells
- Associated with HTLV-L; most commonly seen in Japan and the Caribbean
- Clinical features include rash (skin infiltration), generalized lymphadenopathy with hepatosplenomegaly, and lytic (punched-out) bone lesions with hypercalcemia.

D. Mycosis fungoides

- Neoplastic proliferation of mature CD4⁺ T cells
- Infiltrate the skin, producing localized skin rash, plaques, and nodules.
- Aggregates of neoplastic cells in the epidermis are called Pautrier micro abscesses.
- Cells can spread to involve the blood, producing Sezary syndrome. (Sezary cells are seen on blood smear)

Chronic myeloproliferative disorders

- Hyper-proliferation of neoplastic myeloid progenitors
 - Elevation in one or more formed elements of peripheral blood
 - Associated with mutated tyrosine kinase
- Include: CML, PCV, Primary myelofibrosis, thrombocytopenia

a) Chronic myelogenous leukemia (CML)

- Affects adults
- Associated with BCR-ABL fusion gene ; t(9,22) called Philadelphia chromosome
- Philadelphia chromosome generates constitutive signal that mimics effect of growth factor activation.
- The granulocyte precursors are most affected
- < 5% myeloblasts are seen in blood
- 50 % of individuals enter accelerated phase resembling acute leukemia, 50% of them abruptly without accelerated phase

b) Polycythemia vera (PCV)

- Excessive neoplastic proliferation and maturation of erythroid, granulocytic, and megakaryocytic elements, producing a *panmyelosis*
- Absolute increase in red cell mass, blood volume and viscosity
- Decreased level of erythropoietin in serum
- **JAK2 mutation**
- **Plethoric** congestion of all organs is characteristic
- Enlarged liver and spleen, thrombosis and infarction, hemorrhage
- Progress to myelofibrosis
- **Clinical course**; insidious, plethoric + cyanotic, intense pruritus and peptic ulcer
- GI symptoms, hematemesis, melena, gout and asymptomatic hyperuricemia
- Prognosis, with the help of phlebotomy up to 10 years survival

c) Myeloid Metaplasia with Primary Myelofibrosis

- marrow fibrosis following a brief period in which the peripheral blood white cell and platelet counts are elevated
- extreme hepatosplenomegally-----→inefficient hematopoiesis-----→anemia, thrombocytopenia

- marrow fibrosis is secondary to derangements confined to the hematopoietic cells, particularly the megakaryocytes
- Megakaryocytes release PDGF and TGF β that stimulate marrow fibroblasts to proliferate
- 50% $\rightarrow$ JAK2 mutation

Clinical course: anemia, thrombocytopenia

- Red cell abnormalities include bizarre shapes
- The presence of nucleated red cell precursors and immature white cells is referred to as *leukoerythrocytosis*
- Complication $\rightarrow$ gout

d) Essential thrombocythemia (ET)

- Neoplastic proliferation of mature myeloid cells, especially platelets.
- RBCs and granulocytes are also increased.
- Associated with JAK2 kinase mutation
- Symptoms are related to an increased risk of bleeding and/or thrombosis.
- Rarely progresses to marrow fibrosis or acute leukemia
- No significant risk for hyperuricemia or gout

Histocytic neoplasms

Langerhan cell histiocytoses

- Umbrella designation for a variety of proliferative disorders of histiocytes, or macrophages
- Most histocytic neoplasms are benign and some are malignant
- HX bodies (bribeck granules) in their cytoplasm is characteristic and have tennis racket appearance when seen under electron microscope
- Under light microscope langerhan cell resemble tissue histocytes (macrophages)

Lymphoma

- Neoplastic proliferation of lymphoid cells that forms a mass; may arise in a lymph node or in extra-nodal tissue
- Divided into non-Hodgkin lymphoma (NHL 60%) and Hodgkin lymphoma (HL, 40%)
- NHL is further classified
 1. **Small B cells**—follicular lymphoma, mantle cell lymphoma, marginal zone lymphoma, and small lymphocytic lymphoma (i.e.. CLL cells that involve tissue)
 2. **Intermediate-sized B cells**—Burkitt lymphoma
 3. **Large B cells**—diffuse large B-cell lymphoma

	NHL	HL
Overall frequency	60%	40%
Malignant cells	Lymphoid cells	Reed-Sternberg cells
Composition of mass	lymphoid cells	Predominantly reactive cells
Clinical	Painless lymphadenopathy, usually arises in late adulthood	Painless lymphadenopathy occasionally with 'B' symptoms (like fever), usually arises in young adults
Spread	Diffuse: often extranodal	Contiguous; rarely extra nodal, stereotypical pattern
Staging	Limited importance	Guides therapy; radiation is the mainstay of treatment
Leukemic phase	Occurs	Does not occur

1) Follicular lymphoma

- Neoplastic proliferation of small B cells (CD20+) that form follicle-like nodules
- Clinically presents in late adulthood with painless lymph adenopathy
- Driven by t(14;18)
- *BCL2* on chromosome 18 translocates to the Ig heavy chain locus on chromosome 14,
- Results in overexpression of Bcl2, which inhibits apoptosis
- Progression to diffuse large B-cell lymphoma is an important complication; presents as an enlarging lymph node
- Involvement of visceral sites is uncommon, M=F
- Follicular lymphoma is distinguished from reactive follicular hyperplasia by
 - Disruption of normal lymph node architecture
 - Lack of tingible body macrophages in germinal centers
 - Bcl2 expression in follicles
 - Monoclonality (follicular hyperplasia is polyclonal)

2) Mantle cell lymphoma

- Neoplastic proliferation of small B cells (CD20+) that expands the mantle zone
- Clinically presents in late adulthood with painless lymph adenopathy
- Driven by t(11;14)
- **Clinical course;** fatigue and lymphadenopathy and are found to have generalized disease involving the bone marrow, spleen, liver, and (often) the gastrointestinal tract.
- Are aggressive and incurable, and are associated with a median survival of 3 to 5 years.

3) Marginal zone lymphoma

- Neoplastic proliferation of small B cells that expands the marginal zone
- Associated with chronic inflammatory states such as Hashimoto thyroiditis, Sjogren syndrome, and *H pylori* gastritis
- MALToma is marginal zone lymphoma in mucosal sites.
 - Gastric MALToma may regress with treatment of *H Pylori*,

4) Burkitt lymphoma

- Neoplastic proliferation of intermediate-sized B cells associated with EBV
 - Classically presents as an extra nodal mass in a child or young adult
 - a high-grade tumor
1. African form usually involves the jaw
 2. Sporadic form usually involves the abdomen.
- Burkitt lymphoma is always associated with translocations involving the *MYC* gene on chromosome 8
 - Overexpression of *c-myc* oncogene promotes cell growth.
 - Characterized by high mitotic index and 'starry-sky' appearance on microscopy

5) Diffuse large B-cell lymphoma

- Neoplastic proliferation of large B cells that grow diffusely in sheets
- Most common form of NHL in adults
- t(14;18) translocation involving the *BCL2* gene
- Clinically aggressive (high-grade)
- Arises sporadically or from transformation of a low-grade lymphoma (e.g., follicular lymphoma)
- In late adulthood as an enlarging lymph node or an extra nodal mass (can arise at any age)

- Extra-nodal presentations are common, frequent extra-nodal sites are GIT and the brain.
- Are aggressive tumors that are rapidly fatal if untreated
- With intensive combination chemotherapy complete remission can be achieved in 60%-80% of the patients; of these, approximately 50% remain free of disease for several years and are often cured

HODGKIN LYMPHOMA (HL)

- HL encompasses a group of neoplasms that arise almost invariably in a single lymph node or chain of lymph nodes and spread to the anatomically contiguous nodes.
- It is a tumor of B-cell origin
- RS cells secrete cytokines, Variants of RS - cells are Hodgkin - cell, RS - cell, L&H - cell and lacunar cell
- Attract reactive lymphocytes, plasma cells, macrophages, and eosinophils
- May lead to fibrosis
- Classified as **Classical Hodgkin lymphoma** and **Nodular lymphocyte predominance Hodgkin lymphoma**
- Reactive inflammatory cells make up a bulk of the tumor and form the basis for classification of Classical Hodgkin lymphoma,, Subtypes include

1. Nodular sclerosis

- the most common form(70% of all cases),,
- Involve the lower cervical, supraclavicular, and mediastinal lymph nodes.
- Affects adolescents or young adults
- presence of a particular variant of the RS cell, **the lacunar cell**

2. Mixed cellularity

- the most common form of HL in patients older than the age of 50
- Classic RS cells are plentiful within a distinctive heterogeneous cellular infiltrate
- Mixed cellularity is often associated with abundant eosinophils (RS cells produce IL-5).
- have disseminated disease and systemic manifestations

3. Lymphocyte-rich → Best prognosis

4. Lymphocyte-depletion

- most aggressive out of all types
- Mostly affect the elderly and AIDS patients

Nodular lymphocyte predominance Hodgkin Lymphoma

- comprise about 5% of Hodgkin lymphoma
- is characterized by a large number of small resting lymphocytes admixed with a variable number of benign histiocytes within large, poorly defined nodules
- classic RS cells are extremely difficult to find
- presence of **lymphohistiocytic (L&H)** variant RS cells popcorn like appearance ("popcorn cell")

Clinical course

- a painless enlargement of the lymph nodes
- The enlarged nodes are typically rubbery, discrete and mobile and may achieve a considerable size
- Mediastinal involvement is seen in over half of the patients
- A third of patients with Hodgkin's lymphoma have systemic symptoms

Stage	Distribution of Disease
I	Involvement of a single lymph node region or a single extra lymphatic organ
II	Involvement of ≥ 2 lymph node regions on the same side of the diaphragm alone or with involvement of limited contiguous extra lymphatic organs or tissue
III	Involvement of lymph node regions on both sides of the diaphragm, which may include the spleen, limited contiguous extra lymphatic organ or site, or both
IV	Multiple or disseminated foci of involvement of one or more extra lymphatic organs or tissues with or without lymphatic involvement

Plasma Cell Neoplasm's and related Disorders multiple myeloma

Multiple Myeloma and Related Plasma Cell Disorders

- common feature that is shared among multiple myeloma and the plasma cell dyscrasias is that all originate from a clone of B cells that differentiates into plasma cells and secretes a single complete or partial immunoglobulin
- The disorders are also called Monoclonal gammopathies/ paraproteinemias / dysproteinemias] describe the abnormal Igs

- **M component** may be indicative of an overt B-cell malignancy, they are fairly common in otherwise normal elderly persons
- Increase in serum levels of a single homogeneous immunoglobulin or its fragments
- The plasma cell dyscrasias can be divided into six major variants

I. MULTIPLE MYELOMA

- Malignant proliferation of plasma cells in the bone marrow
- **Most common** primary malignancy of bone
- Usually associated with *multifocal lytic lesions throughout the skeletal system*
- In approximately 60% of patients, the M component is IgG; in 20% to 25%, IgA; rarely, it is IgM, IgD, or IgE
- Readily excreted in urine, where they are termed **Bence Jones proteins**
- Myeloma nephrosis, hypercalcemia due to bone resorption (metastatic calcification)
- Peak age incidence of MM is 50-60 yrs
- Bone pain, Recurrent infections, Hyper-viscosity syndrome, Renal insufficiency and Amyloidosis

II. Localized Plasmacytoma

- Solitary lesions involving the skeleton or the soft tissues.
- Extra osseous lesions occur mainly in the **upper respiratory tract**
- Modestly elevated M proteins are demonstrable in some of these patients.

III. Lymphoplasmocytic Lymphoma

- mixed proliferation of B cells

- Commonly involves multiple lymph nodes, bone marrow, and spleen at time of presentation.
- the tumor produces an M component,
- Unlike myeloma, this disease *does not produce lytic bone lesions*.

IV. Heavy-Chain Disease

- A group of proliferations in which only heavy chains are produced, most commonly IgA.
- IgA heavy-chain disease shows a predilection for the lymphoid tissues where IgA is normally produced

V. Primary or Immunocyte-Associated Amyloidosis

- The amyloid deposits (of AL type) consist of partially degraded light chains.

VI. Monoclonal Gammopathy of Undetermined Significance. (MGUS)

- Increased serum protein with M spike
- no lytic bone lesions, hypercalcemia, AL amyloid, or Bence Jones proteinuria
- Common in elderly (seen in 5% of 70-year-old individuals);
- 1% of patients with MGUS develop multiple myeloma each year.

VII. W ALDENSTROM M ACRO G LOB U LIN EM IA

- B-cell lymphoma with monoclonal IgM production
- Generalized lymphadenopathy, lytic bone lesions are absent.
- Increased serum protein with M spike (comprised of IgM)
- Visual and neurologic deficits, IgM causes serum hyper-viscosity.
- Bleeding—Viscous serum results in defective platelet aggregation.

- Acute complications are treated with plasmapheresis, which removes IgM from the serum

PHARMACOLOGY OF IMMUNOLOGY

Anti-anemic drugs

- Anemia is a common clinical condition that is caused by an acquired or hereditary abnormality of RBCs or its precursor or it may be a manifestation of and underlying non hematologic disorder.
- Anemia is defined as a decrease in the circulating RBC mass; the usual criteria are a HB of less than 12G/dl in women and less than 14 G/dl in men.
- Patients with an Hb less than 7G/dl will have symptoms of tissue hypoxia fatigue, headache, dyspnea, pallor, angina, tachycardia, visual impairment, syncope, lymphadenopathy, hepatic or splenic enlargement, vasodilation, bone tenderness, blood loss in faeces, neurologic symptoms.

▪ **IRON PREPARATION**

- Oral iron
 1. Ferrous sulfate-325mg-65mg elementaliron
 2. Ferrous gluconate-320mg-37mg elemental iron
 3. Ferrous fumarate-325mg-106mg elemental iron
- Parenteral iron; iron dextran 50mg elemental iron/ml

▪ Pharmacokinetics

- **Absorption:** iron (Fe^{++}) absorb from duodenum and upper jejunum $\rightarrow$ Fe^{+++} (ferric) in the intestinal mucosal cell.
- Intestinal epithelial cells actively absorb inorganic iron and heme iron (H).
- **Distribution:** Ferrous iron that is absorbed or released from absorbed heme iron in the intestine is actively transported into the blood or complexed with Apo ferritin (AF) and stored as ferritin.

- In the blood, iron is transported by transferrin (Tf) to erythroid precursors in the bone marrow for synthesis of hemoglobin (Hgb) or to hepatocytes for storage as ferritin.
 - The transferrin-iron complexes bind to transferrin receptors (TfR) in erythroid precursors and hepatocytes and are internalized.
 - After release of iron, the TfR-Tf complex is recycled to the plasma membrane and Tf is released.
 - About 10-20% of dietary iron is absorbed.
 - **Storage:** ferritin and hemosiderin form in the mucosal cells, liver, spleen, and bone marrow.
 - **Elimination:** minimal amount (about 1 mg/day) are lost in sweat, saliva, and in exfoliated skin and intestinal mucosal cell.
- Iron deficiency is commonly seen in premature infants, children during rapid growth, pregnant, lactating mothers, and patients with small bowel disease that results in generalized malabsorption.
 - as iron deficiency develops storage iron decreases and then disappears, serum ferritin decreases, then serum iron decrease, TIBC (total iron binding capacity) increase.
 - **Oral iron therapy**
 - 200-400 mg oral elemental iron daily should be given to correct anemia (25% absorbed so 50-100 mg iron can be incorporated in Hb).
 - Treatment should continue for 3-6 months, this not only correct the anemia but will replenish iron stores. (Hb should reach normal level in 1-3 months).

- Adverse effects: - nausea, epigastric discomfort, abdominal cramps, constipation, diarrhea, black stool. These can be minimized by lowering the daily dose or by taking iron tablets immediately after or with meals.

▪ Parenteral iron therapy

- Should be reserved for patients who are unable to tolerate or absorb iron (pts with post gastrectomy, previous small bowel resection, malabsorption syndrome).
- Pts with extensive chronic blood loss
- Iron dextran (ferric hydroxide dextran), 50 mg elemental iron/ml, by IM or IV infusion in 1-2 hours
- Adverse effects: - local pain, tissue staining (brown discoloration of tissues overlying the injected site), headache, fever, arthralgia, nausea, vomiting, bronchospasm, urticaria, anaphylaxis, and death.

▪ Acute iron toxicity

- Seen in young children who have ingested iron tablets causing necrotizing gastroenteritis, vomiting, abdominal pain, bloody diarrhea, dyspnea, metabolic acidosis, coma and death.
- Treatment
 - . Whole bowel irrigation
 - . Deferoxamine, potent iron chelating, it binds iron that has already been absorbed and to promote its excretion in urine and feces.
 - . Supportive therapy for GIT bleeding, metabolic acidosis and shock

▪ Chronic iron toxicity

- Known as hemochromatosis, when excess iron is deposited in heart, liver, pancreas, and other organs causing organ failure and death.
- It occurs in pts with inherited hemochromatosis (excessive iron absorption in pts who receive many red cell transfusions for long period).
- Treatment
 - . Intermittent phlebotomy, 1 unit of blood removed /weekly
 - . Iron chelating agent (deferoxamine IV).
 - . Deferoxamine: - is poorly absorbed when given orally and is metabolized and excreted in urine (turn urine color orange red).

▪ Adverse effects

- Rapid IV administration → hypotension
- Idiosyncratic response such as flushing, erythema, intestinal irritation, urticaria
- Acute respiratory distress syndrome occur if IV infusion lasting longer than 24 hours
- Neurotoxicity after long term therapy of iron overload condition

▪ VITAMIN B12

- Deficiency of either vitamin B12 or folic acid manifests as megaloblastic anemia.
- Deficiency of vitamin B12 leads to anemia, GI symptoms, and neurologic abnormality.
- Pharmacokinetics
 - . It is produced only by bacteria.
 - . It is absorbed from the GIT in the presence of intrinsic factor, a product of the parietal cells of the stomach.
 - . Plasma transport is accomplished by binding to trans-cobalamin 2.

. Vitamin B12 is stored in the liver in large amounts; a normal individual has enough to last 5 yrs.

. It's available in 2 forms: - cyanocobalamin and hydroxycobalamin (has longer circulating half-life).

- Pharmacodynamics

. Vitamin B12 is essential in 2 reactions:-

1. Conversion of methyl malonyl-coenzyme A to succinyl-co A and conversion of homocysteine to methionine.
2. The second reaction is linked to folic acid metabolism and synthesis of deoxythymidylate (dTTP), a precursor required for DNA synthesis.

. In Vitamin B12 deficiency, folates accumulate as N-methyl tetrahydrofolate, the supply of tetrahydrofolate is depleted and the production of RBCs slows.

. Administration of folic acid to patients with Vitamin B12 deficiency helps refill the tetrahydrofolate pool and partially or fully corrects anemia.

. However exogenous folic acid does not correct the neurologic defects of vitamin B12 deficiency.

▪ Clinical uses

- Treatment of naturally pernicious anemia
- Anemia caused by gastric resection
- Replacement of Vitamin B12 using parenteral therapy.

▪ No significant toxicity occurred.

FOLIC ACID

- Its deficiency during pregnancy increases the risk of neural tube defects in the fetus.
- Pharmacokinetics: - folic acid is readily absorbed from GIT. Only modest amount are stored in the body.
- Pharmacodynamics: - folic acid is converted to tetrahydrofolate by the action of dihydrofolate reductase.
- Readily dividing cells are highly sensitive to folic acid deficiency, for this reason ant folate drugs are useful in treatment of various infections and cancers.
- Folic acid supplementation is recommended prior to and during pregnancy.
- Has no toxicity.
- It will correct the anemia but not the neurologic deficits of Vitamin B12 deficiency.

▪ Erythropoietin

- Is produced by the kidney
- Reduction in its synthesis is responsible for anemia of renal failure.
- Is used for anemia associated with renal failure and sometime effective for patients with other forms of anemia. E.g primary bone marrow disorder or anemia secondary to cancer chemotherapy or HIV treatment, bone marrow transplantation, AIDS or cancer.
- Toxicity: - thrombosis, cardiovascular events when used along with other erythropoietic agents.

ANTICOAGULANTS, THROMBOLYTIC, ANTIPLATELETS DRUGS

▪ ANTICOAGULANTS

- Heparin, hirudin, warfarin, argatroban, drotrecogin alfa

▪ THROMBOLYTICS

- Streptokinase, tissue plasminogen, activators(tPAs), urokinase

▪ ANTIPLATELET DRUGS

- Abciximab, acetylsalicylic acid, dipyrid amole, ticlopidine

▪ ANTIDOTES

- Phytonadione(vitamin k1), protamine sulfate, aminocaproic acid(EACA)

▪ These drugs are used to treat strokes, myocardial infarctions, pulmonary embolisms, disseminated intravascular coagulation (DIC) and deep vein thrombosis (DVT).

▪ **Fibrinolytic agents:** - degrade fibrinogen/fibrin

- Eliminate formed clots

▪ **Anticoagulants:** - inhibit clotting mechanisms

- Prevent progression of thrombosis

▪ **Antiplatelet drugs:** - interfere either with platelet adhesion or aggregation

- Prevent initial clot formation

- **ANTICOAGULANT DRUGS**

- **MOA**

1. Activation of anticlotting factors (esp antithrombin 3) e.g heparin, dalteparin, enoxaparin, fondaparinux

- HEPARIN**

- . Most commonly given anticoagulant for short term use (>12 million patients/year).
 - . It is a complex mixture of mucopolysaccharide
 - . Isolated from bovine lung or porcine intestine.
 - . Normally found in mast cells – unknown physiologic role.
 - . Strongly acidic.

- DALTEPARIN, ENOXAPARIN**

- . Low MW fragments of HEPARIN

- FONDAPARINUX**

- . Synthetic formulation of the key pentasaccharide that appears to the active component in all heparins.

- MOA**

- Bind to antithrombin 3, a protease inhibitor that complexes with activated clotting factors 2, 4, 5, and 6 (i.e. heparins are indirect thrombin inhibitors)
 - Coat blood vessel wall (prevents platelet binding or causes permeabilization).

- PHARMACOKINETICS**

- Activity is standardized by bioassay
- Must be given IV (IM administration causes hematomas)

- Rapid effect within minutes; instantaneously in vitro
- Metabolized in liver; excreted in urine
- Heparin doses adjusted to double partial thromboplastin time.

THERAPEUTIC INDICATIONS

- Effective anticoagulant in vitro
- For massive deep vein thrombosis
- Pulmonary infarct
- Post-operative acute MI (except brain, spinalcord or eye)

ADVERSE EFFECTS

- Hemorrhage (esp. Hemorrhagic stroke)
- Hypersensitivity
- Unique side effect of heparin is a transient thrombocytopenia (HIT)

. In these, heparin should be discontinued and treatment initiated with hirudin or lepirudin, but not warfarin (may exacerbate the prothrombotic state).

2. Direct inhibition of thrombin e.g. hirudin, argatroban, bivalirudin, lepirudin, ximelagatran

MOA

- The approved use of direct thrombin inhibitors is for treatment of HIT.
 - Work by inhibiting fibrin binding to thrombin → interacting with the thrombin active site → inhibiting thrombin activity even in the presence of bound fibrin
 - ARGATROBAN is a synthetic derivative of l-arginine – ARGATROBAN and XIMELAGATRAN bind reversibly to the thrombin active site
 - Advantages over heparin
1. Inhibition of coagulation via a single mechanism of action

2. Actions are independent of antithrombin 3, which means they can reach and inactivate fibrin-bound thrombin inside clots
3. Little effects on platelets or bleeding time
4. Elimination of HIT as a side effect of treatment
5. Not inactivated by platelet or plasma proteins
6. More uniform potency and increased safety
7. Rebound coagulation after discontinuation of the drug is less likely with direct thrombin inhibitors

PHARMACOKINETICS

- Given IV (note: XIMELAGATRAN is the first oral agent in this class → promoted as a replacement for both WARFIN and HEPARIN because of its immediate anticoagulant action)
- Metabolized by hydrolysis in liver, excreted in urine

ADVERSE EFFECTS

- HEMORRHAGE: dose related; increased in patients with renal insufficiency, liver injury, or recent trauma or treatment with other anticoagulants
- 3. Inhibition of synthesis of blood coagulation factor precursors (zymogens) e.g WARFARIN

MOA

- Interfere with the synthesis of vitamin K and thus inhibits activation of vitamin K-dependent clotting factors (2,7,9,10)
- Also decreases the activity of protein C (activated by thrombin)-responsible for some side effects.

PHARMACOKINETICS

- WARFARIN- has a narrow therapeutic index, is nearly completely (99%) bound to plasma albumin, is eliminated by hepatic metabolism (CYP450)
- Warfarin is the prototype for drug-drug interactions
- Rapid complete absorption from GI: peak plasma drug concentration in ≤ 1 hr
- Warfarin is a racemic mixture of R- and S-forms – S is the active enantiomer
- S and R warfarin are metabolized differently ; S is metabolized primarily by CYP2C9 and R by CYP1A2, CYP2C19 and CYP3A4
- Several genes play a role in WARFARIN metabolism
- Polymorphisms in CYP2C9 cause about 30% of patients to be slow metabolizers (causes increase serum concentration)
- Polymorphisms in the vitamin K epoxide reductase multi-protein complex (VKOR) can confer resistance (decrease serum concentration)
- No in vitro effect
- Readily crosses placenta

THERAPEUTIC INDICATIONS

- Long term oral treatment of deep vein thrombosis
- For 2-6 months following myocardial infarction
- Atrial fibrillation

ADVERSE EFFECTS

- Hemorrhage
- Flatulence and diarrhea are common

- Decreased protein C causes cutaneous necrosis caused during first weeks of treatment; rarely can progress to infarction (venous thrombosis) in fatty tissues, intestine and extremities
- Bone defects (chondrodysplasia punctata) and hemorrhagic disorders in infants born to mothers taking the drug during first trimester of pregnancy

→ **ABSOLUTELY CONTRAINDICATED IN PREGNANCY**

4. Activated protein c i.e., DROTRECOGIN ALFA

- Is approved for use in disseminated intravascular coagulation or DIC (fatal complication of septic shock). The use of activated protein C as a drug occurred as a result of a change in our understanding of the pathophysiology of sepsis, particularly the intricate interplay between activation of coagulation and inflammation.

▪ **THROMBOLITIC DRUGS**

Alteplase, reteplase, tenecteplase, anistreplase, streptokinase, urokinase

- Dissolve existing life threatening thrombi
- Activate plasminogen to plasmin → hydrolysis of fibrin and several other coagulation factors
- Short activation time and short half lives
- Recommended for patients with
 - . Recent acute MI
 - . Extensive pulmonary emboli
 - . Severe deep vein thrombosis
 - . Thromboembolic stroke (tPAs only)
- Common contraindications

. Cause bleeding (esp. hemorrhagic stroke) - can be antagonized by AMINOCAPROIC ACID (for tPAs) or APROTININ (for STREPTOKINASE)

. Not for use in recent surgery, pregnancy, acute pericarditis, hypertension pts, GI bleeding..

. Its expensive (esp. tPAs)

1. ANISTREPLASE, STREPTOKINASE

MOA

- Binds to and induces a conformational change in plasminogen resulting in exposure of the active site and conversion to plasmin (streptokinase is not intrinsically active)
- Anistreplase is an inactive complex of streptokinase and human lys-plasminogen more convenient (shorter infusion time) but far more expensive with an increased tendency for systemic thrombolysis

ADVERSE EFFECTS

- May evoke allergic hypersensitivity reactions, fever, and anaphylaxis

2. UROKINASE

MOA

- Kidney enzyme that directly converts plasminogen to active plasmin
- Primarily indicated only for patients allergic to streptokinase

ADVERSE EFFECTS

- Febrile episodes are common but infusion reactions and hypersensitivity (anaphylaxis) are rare

3. TISSUE PLASMINOGEN ACTIVATORS (tPAs):ALTEPLASE, RETEPLASE, TENECTEPLASE

MOA

- Causes selective activation of fibrin bound plasminogen
- Poor plasminogen activator in the absence of fibrin i.e. theoretically confines fibrinolysis to the formed thrombus and decreases systemic activation

▪ ANTIPLATELET DRUGS

ASA, ABCIXIMAB, CLOPIDOGREL, DIPYRIDAMOLE, EPTIFIBATIDE, TICLOPIDINE, TIROFIBAN

- **MOA: Inhibit platelet adhesion and aggregation by;-**
 1. Inhibiting cyclooxygenase: e.g. ASA
 2. Blocking glycoprotein 2b/3a receptor: e.g. ABCIXIMAB, EPTIFIBATIDE
 3. Inhibiting the binding of fibrinogen to activated platelets: e.g. CLOPIDOGREL, TICLOPIDINE
 4. Inhibiting cyclic nucleotide phosphodiesterase: e.g DIPYRIDAMOLE

It is with great regret that we inform you that the following topics are not included in this preparatory module.

- Cytokine mediated diseases and therapy
- Immune tolerance
- Immunity to microbial and parasitic infection
- Vaccine preventable disease
- Immunotherapy

Please complete your reading from slides or any other reference materials. It is our sincere hope that you find this module quite helpful. Good luck in your preparation and God-speed.

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